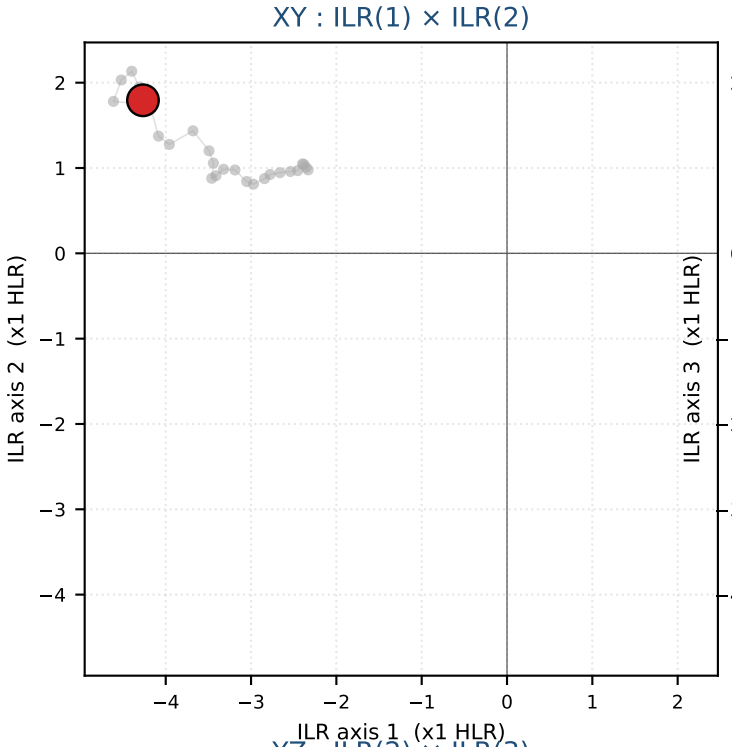
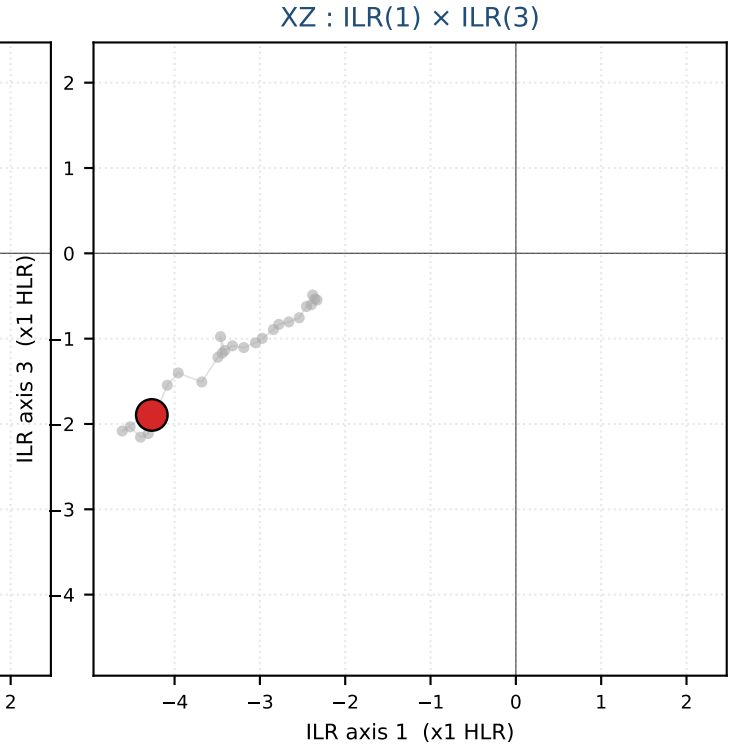
STAGE 1 v4 | t=0 | 2000 | D=8 T=26 | level 1 first-principles

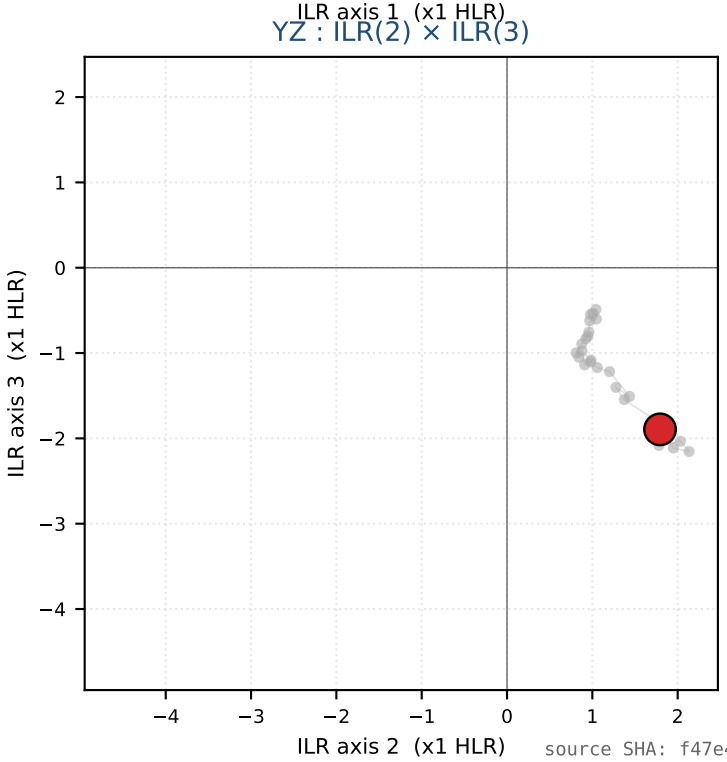
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

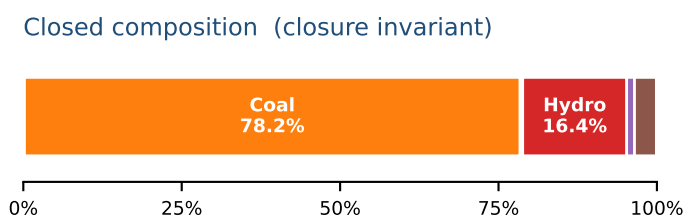




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0019	-1.229
Coal	0.7821	+4.805
Gas	0.0043	-0.409
Hydro	0.1641	+3.243
Nuclear	0.0123	+0.657
Other Fossil	0.0349	+1.695
Solar	0.0000	-6.073
Wind	0.0004	-2.689





closure: Σ = 1.000000 OK

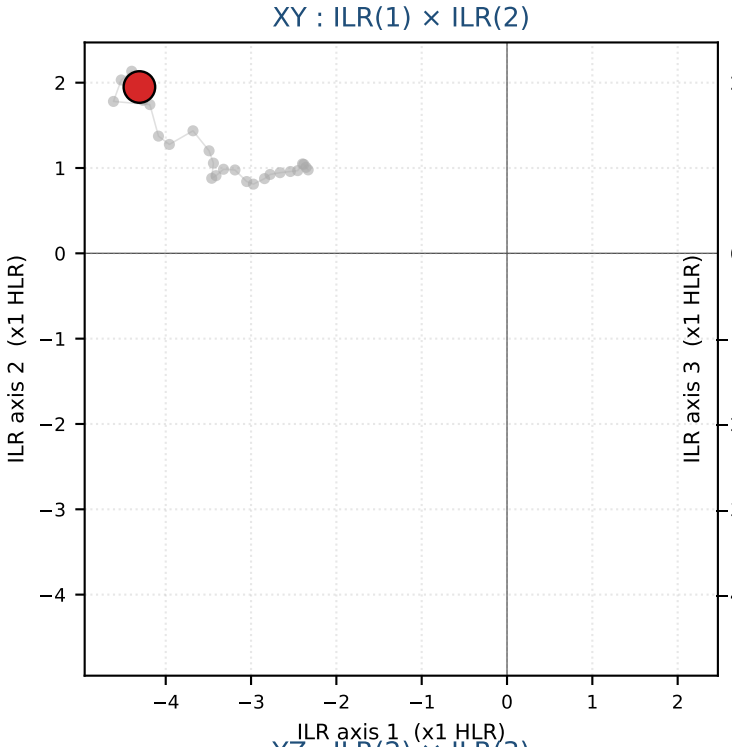
display factor: x1 (raw HLR x 1)

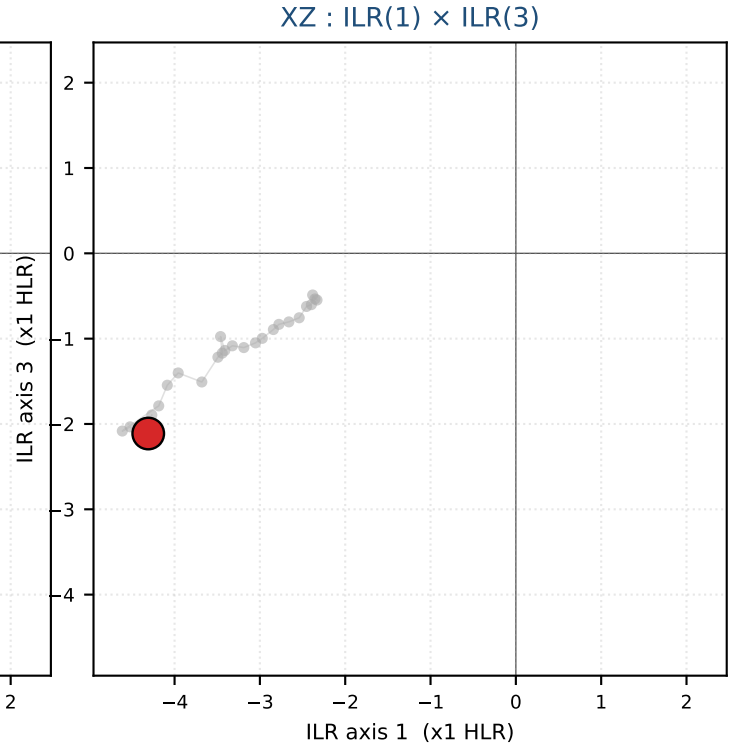
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=1 | 2001 | D=8 T=26 | level 1 first-principles

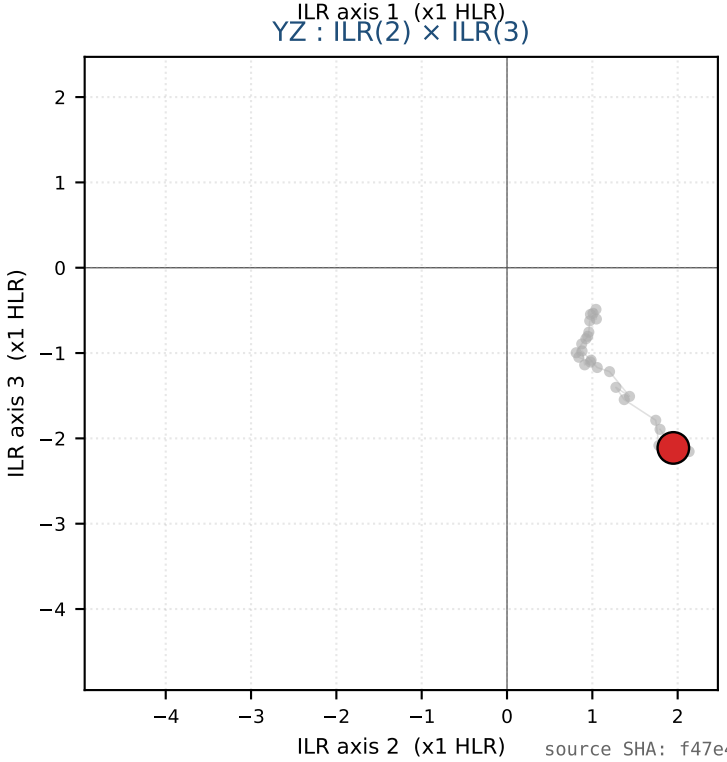
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

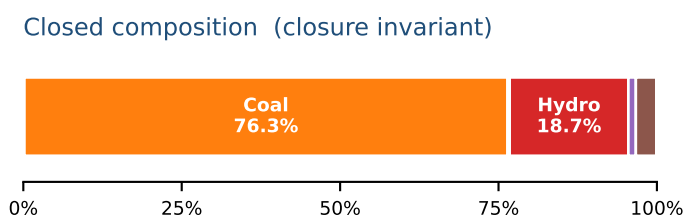




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0017	-1.326
Coal	0.7625	+4.768
Gas	0.0033	-0.666
Hydro	0.1874	+3.364
Nuclear	0.0118	+0.599
Other Fossil	0.0328	+1.620
Solar	0.0000	-5.768
Wind	0.0005	-2.590





closure: Σ = 1.000000 OK

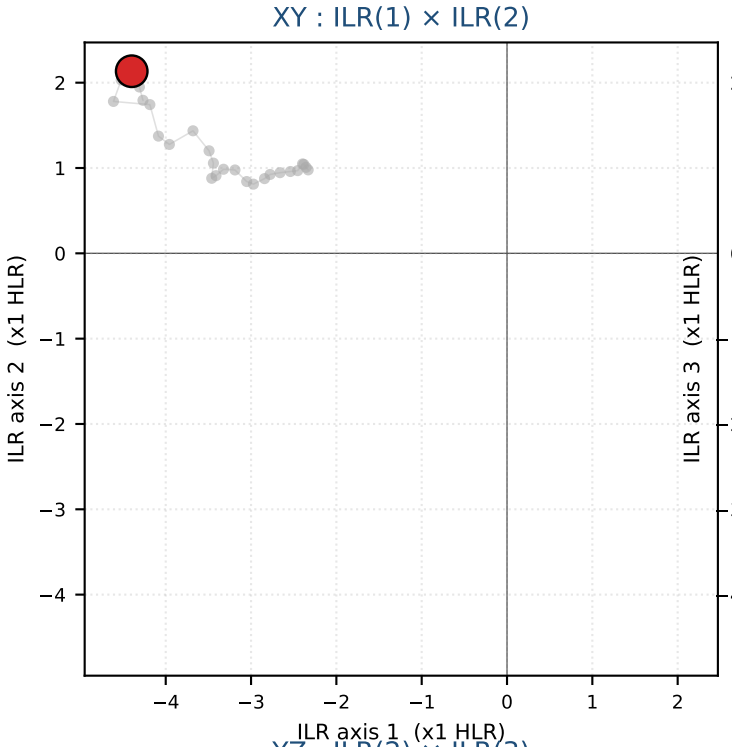
display factor: x1 (raw HLR x 1)

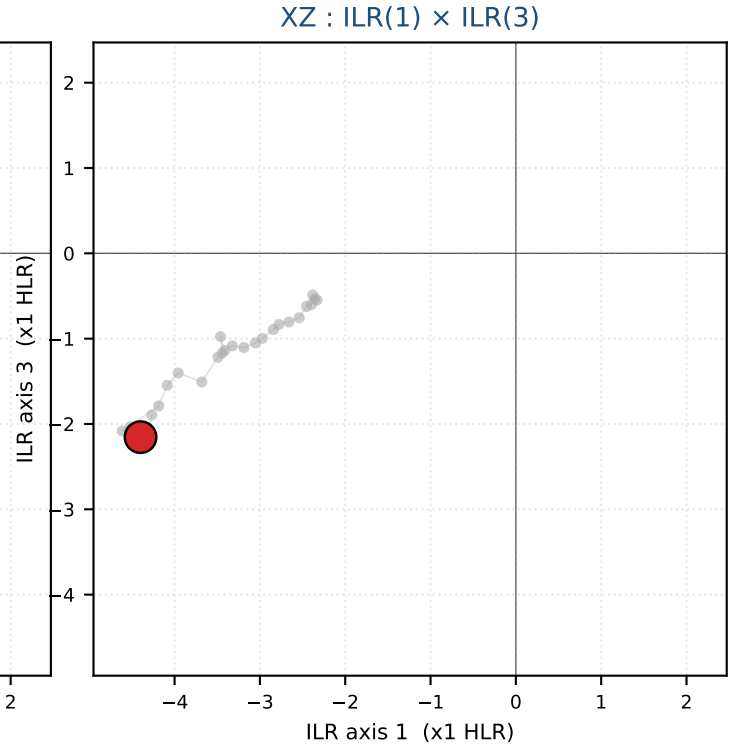
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=2 | 2002 | D=8 T=26 | level 1 first-principles

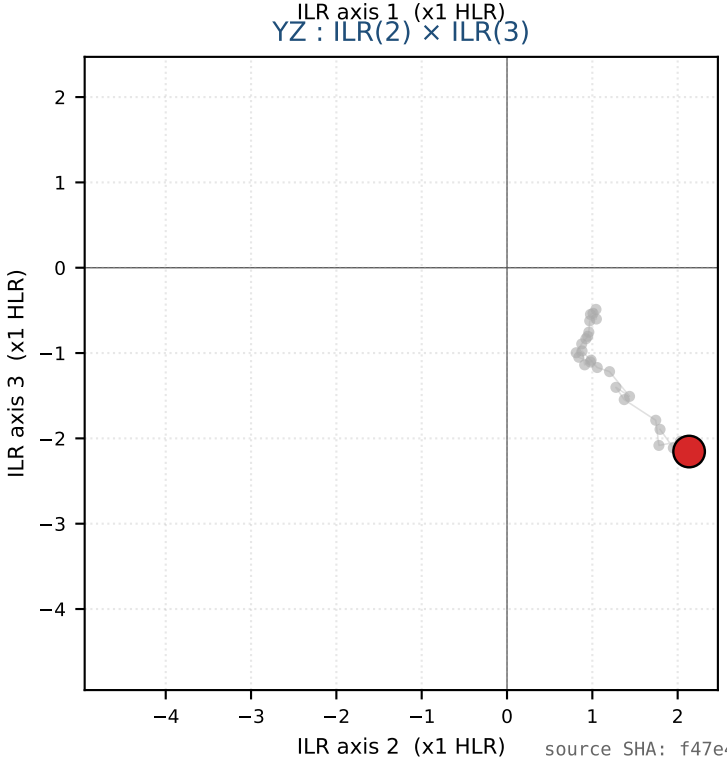
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

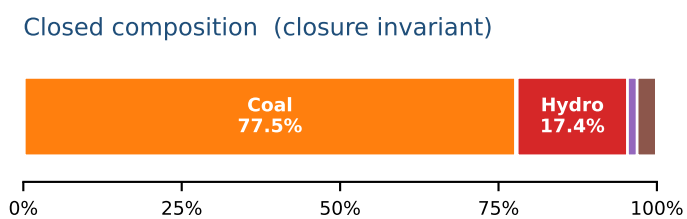




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0015	-1.460
Coal	0.7754	+4.761
Gas	0.0025	-0.963
Hydro	0.1741	+3.267
Nuclear	0.0152	+0.828
Other Fossil	0.0307	+1.530
Solar	0.0000	-5.392
Wind	0.0005	-2.570





closure: Σ = 1.000000 OK

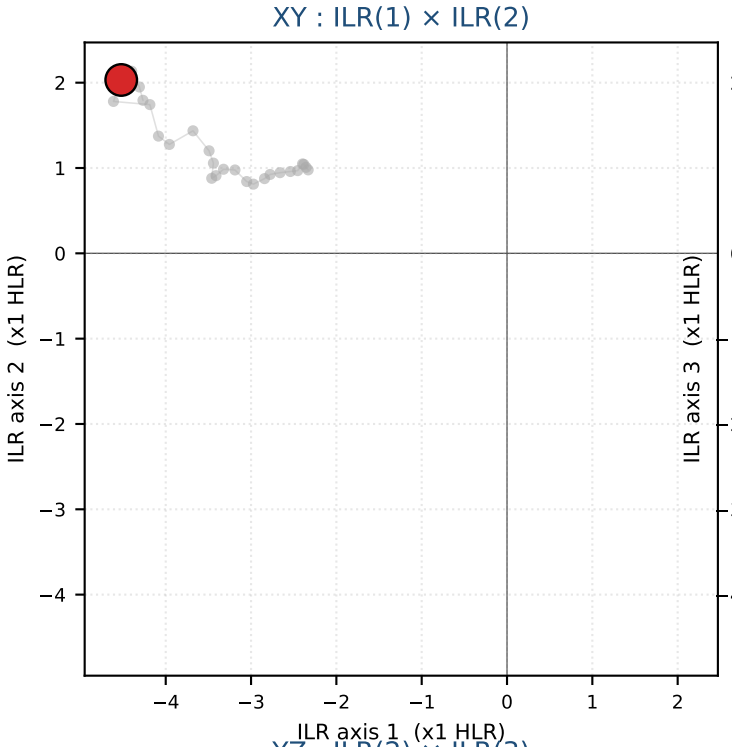
display factor: x1 (raw HLR x 1)

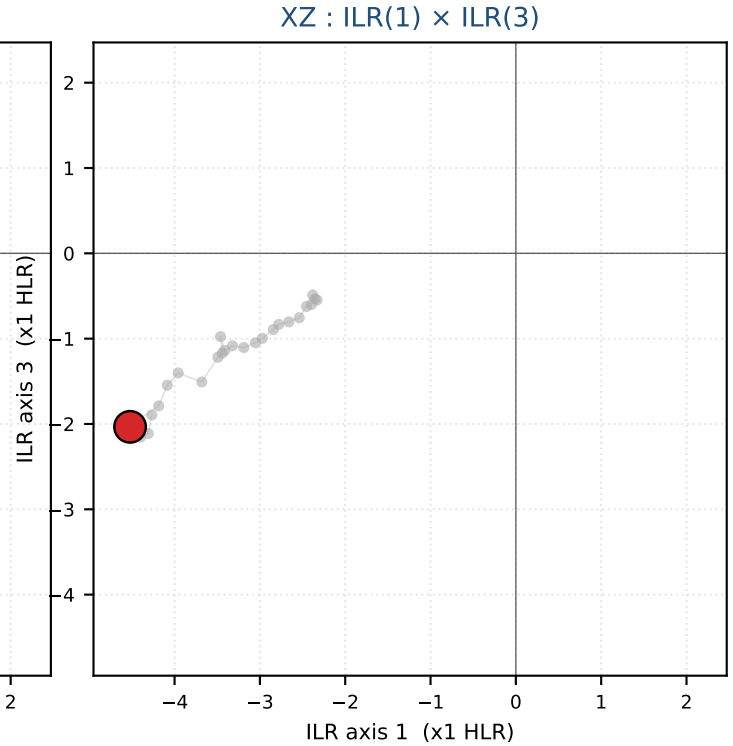
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=3 | 2003 | D=8 T=26 | level 1 first-principles

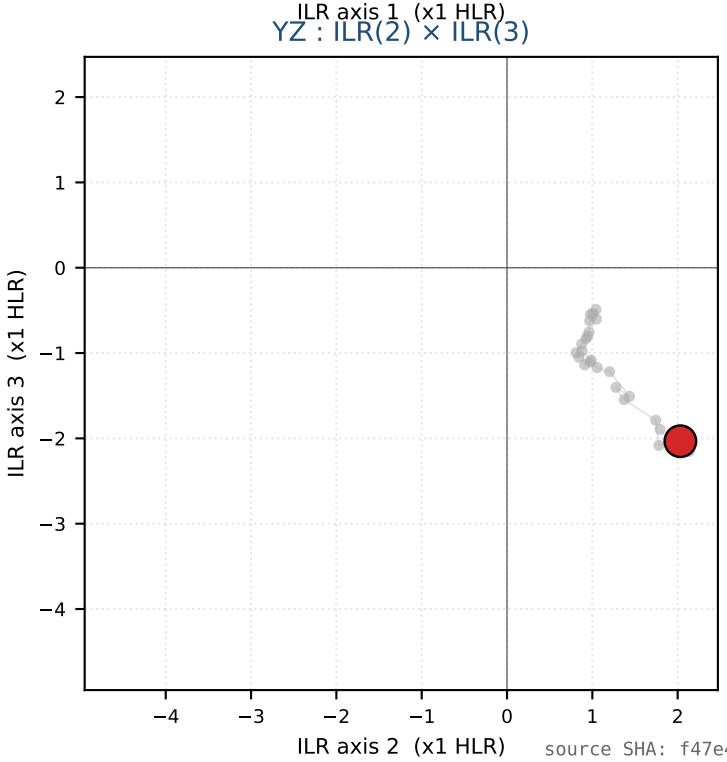
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

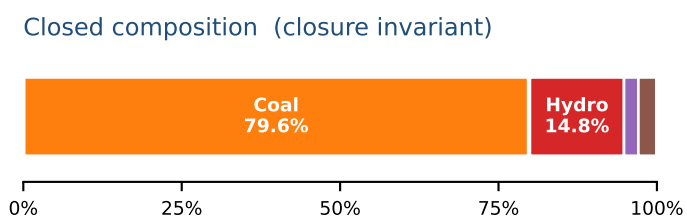




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0013	-1.631
Coal	0.7956	+4.763
Gas	0.0027	-0.922
Hydro	0.1485	+3.084
Nuclear	0.0227	+1.206
Other Fossil	0.0287	+1.441
Solar	0.0000	-5.377
Wind	0.0005	-2.563





closure: Σ = 1.000000 OK

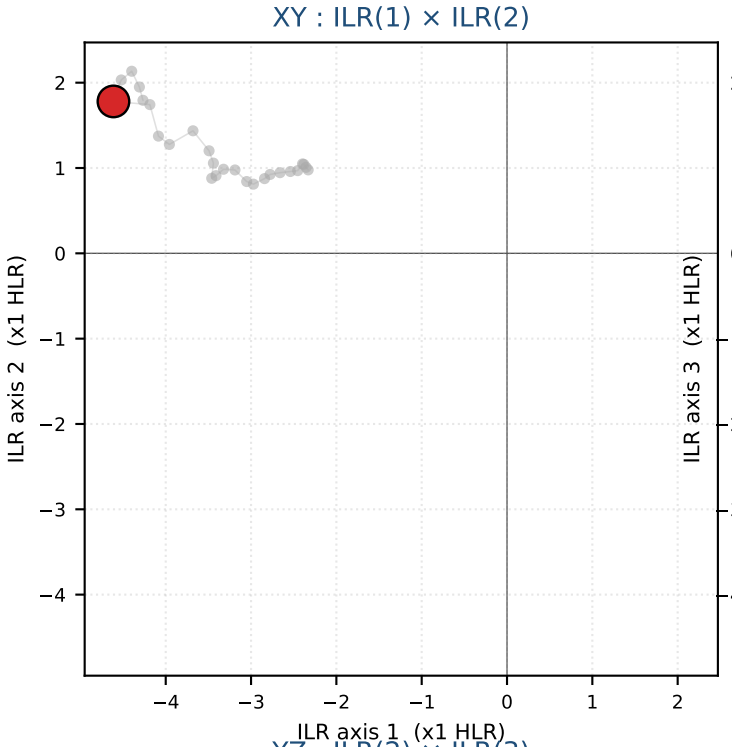
display factor: x1 (raw HLR x 1)

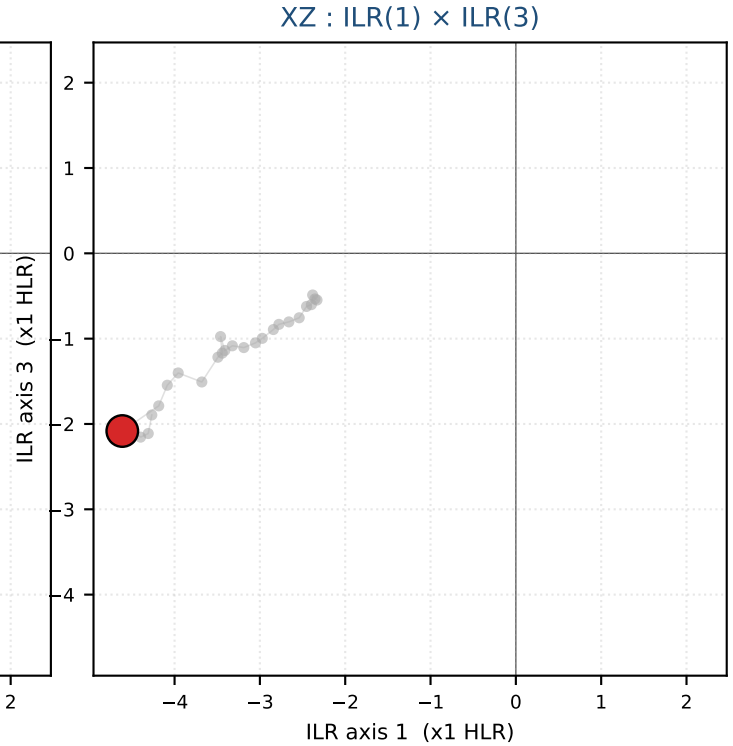
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=4 | 2004 | D=8 T=26 | level 1 first-principles

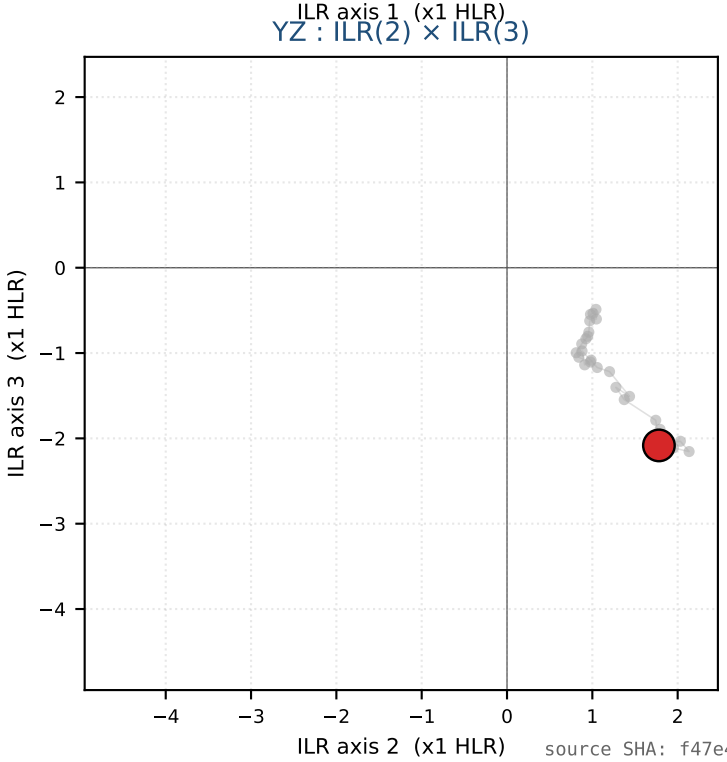
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

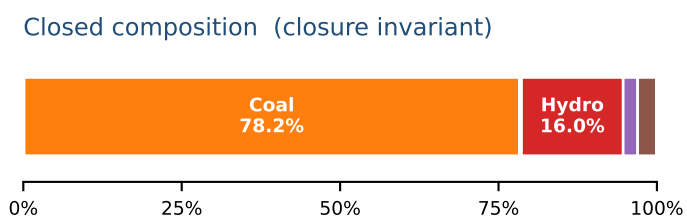




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0011	-1.832
Coal	0.7817	+4.691
Gas	0.0034	-0.751
Hydro	0.1605	+3.108
Nuclear	0.0229	+1.161
Other Fossil	0.0297	+1.422
Solar	0.0000	-5.286
Wind	0.0006	-2.513





closure: Σ = 1.000000 OK

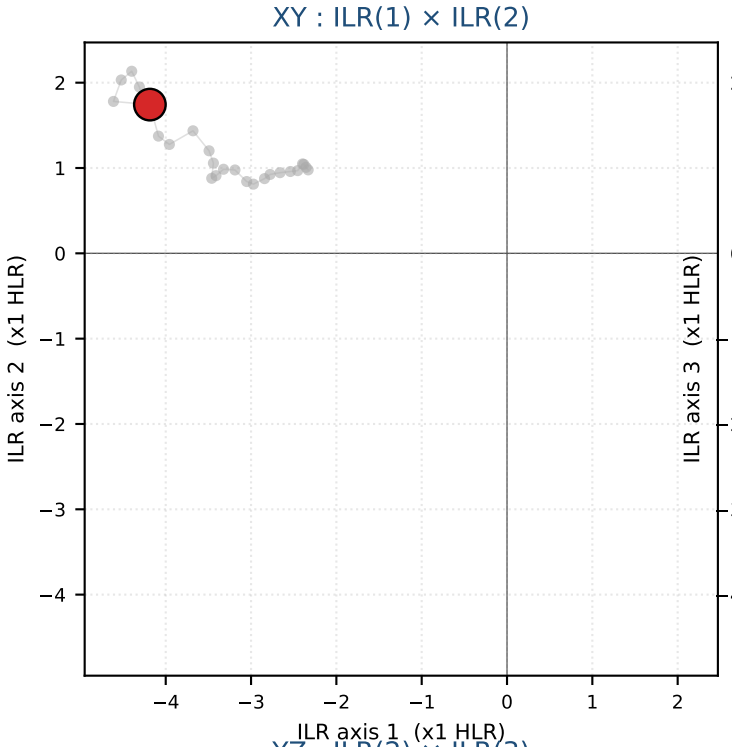
display factor: x1 (raw HLR x 1)

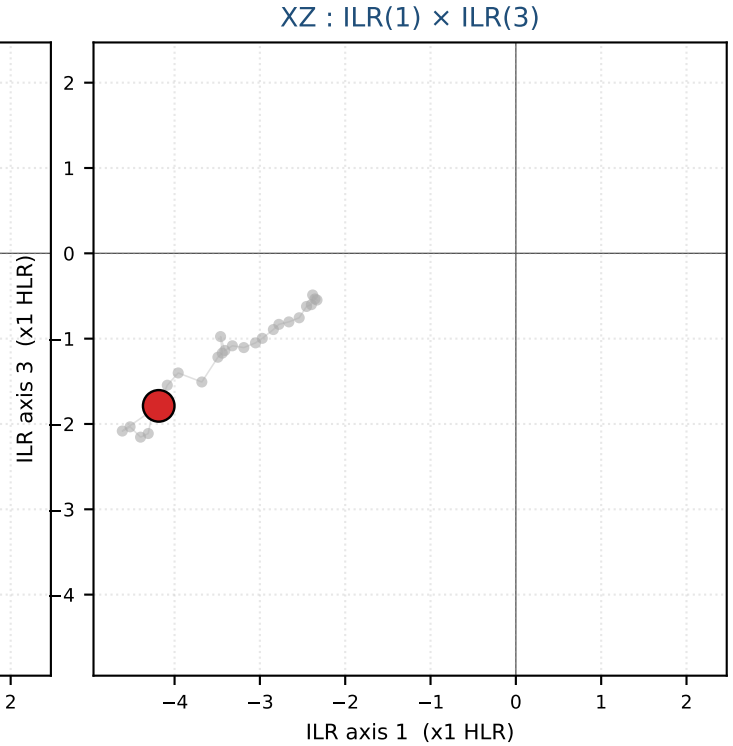
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=5 | 2005 | D=8 T=26 | level 1 first-principles

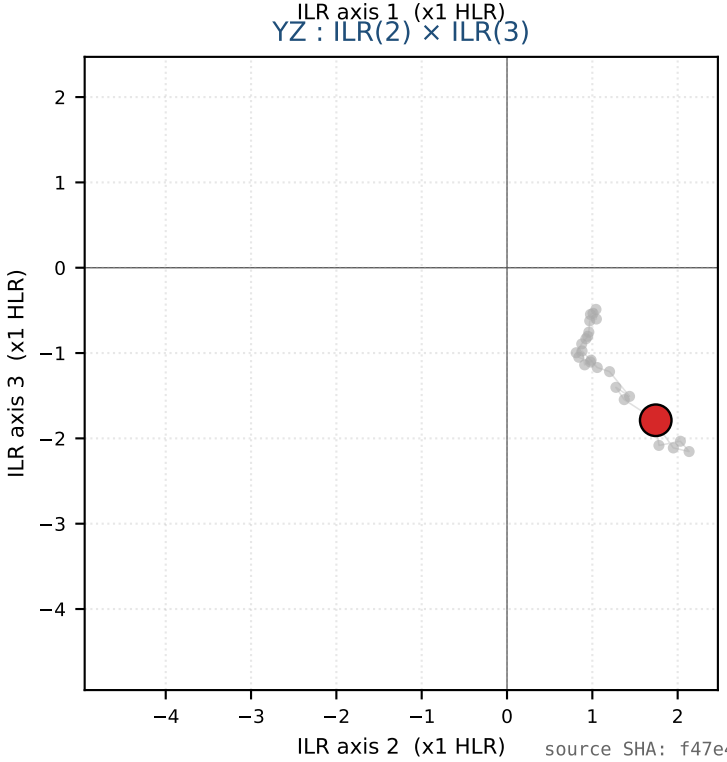
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

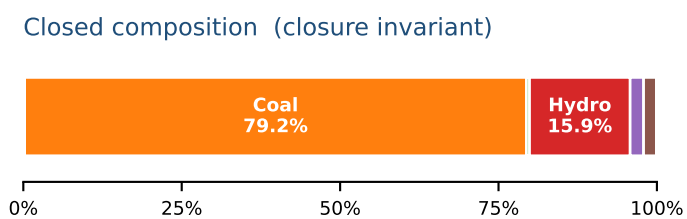




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0021	-1.301
Coal	0.7920	+4.619
Gas	0.0049	-0.476
Hydro	0.1588	+3.012
Nuclear	0.0212	+1.000
Other Fossil	0.0202	+0.950
Solar	0.0000	-5.498
Wind	0.0008	-2.305





closure: Σ = 1.000000 OK

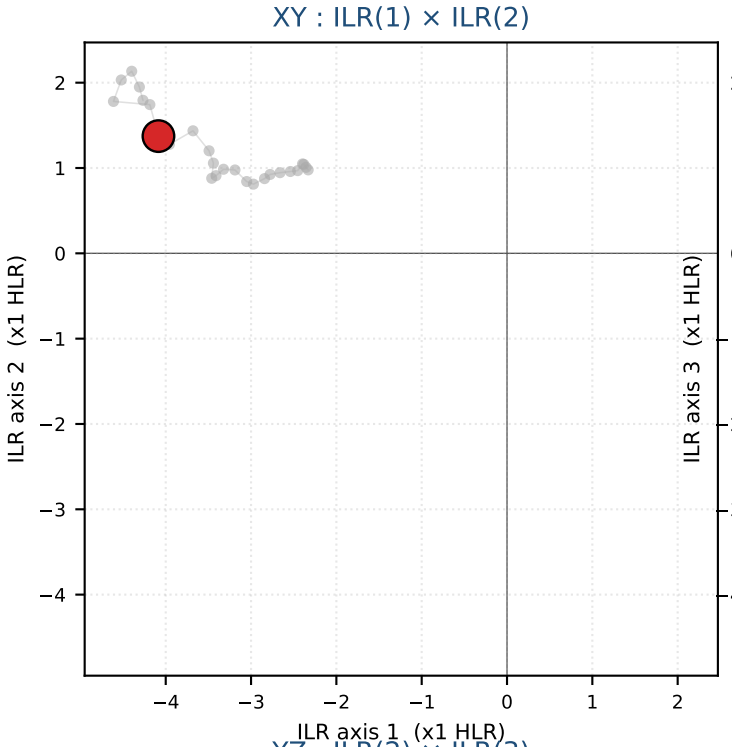
display factor: x1 (raw HLR x 1)

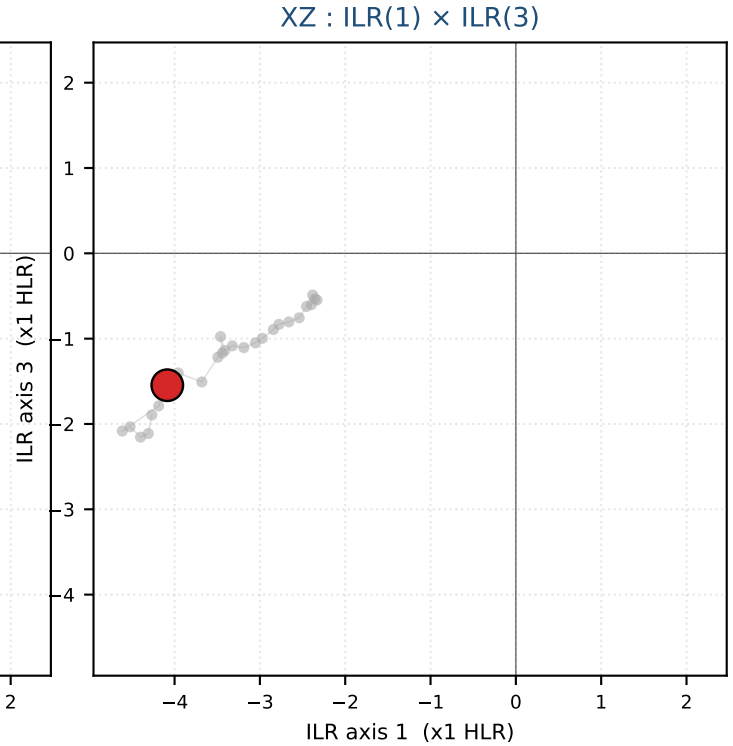
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=6 | 2006 | D=8 T=26 | level 1 first-principles

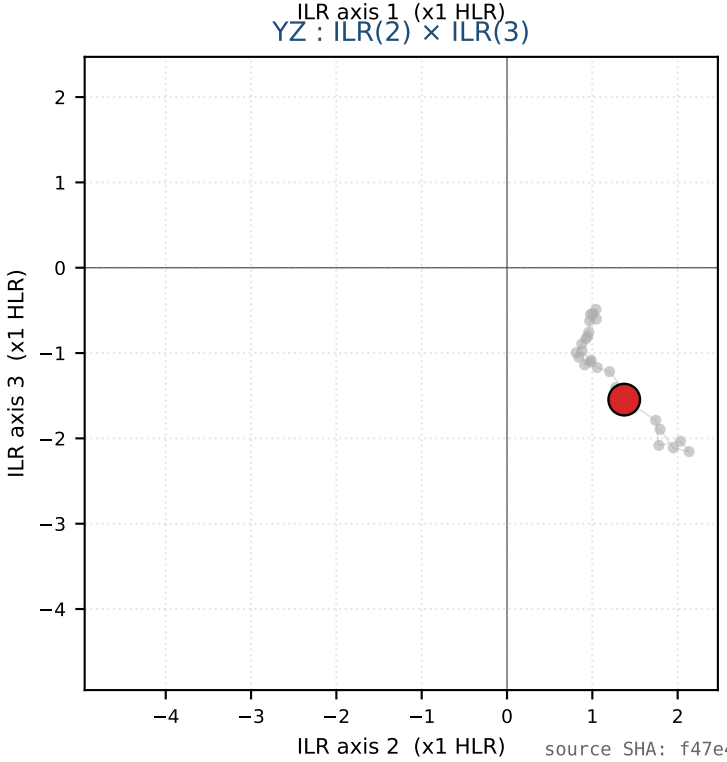
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

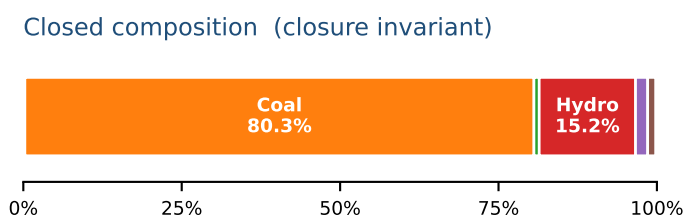




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0025	-1.223
Coal	0.8034	+4.555
Gas	0.0083	-0.017
Hydro	0.1521	+2.890
Nuclear	0.0191	+0.817
Other Fossil	0.0132	+0.448
Solar	0.0000	-5.595
Wind	0.0013	-1.876





closure: Σ = 1.000000 OK

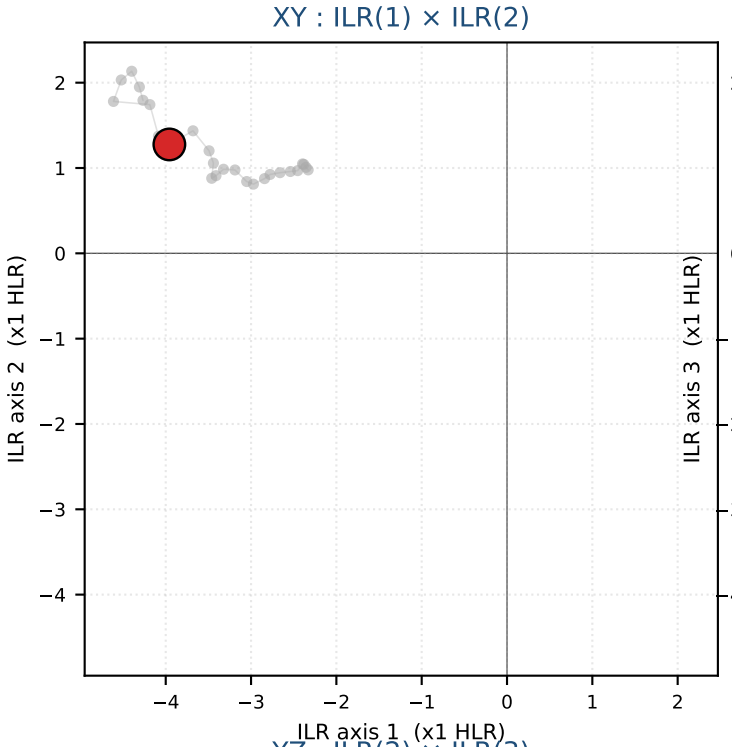
display factor: x1 (raw HLR x 1)

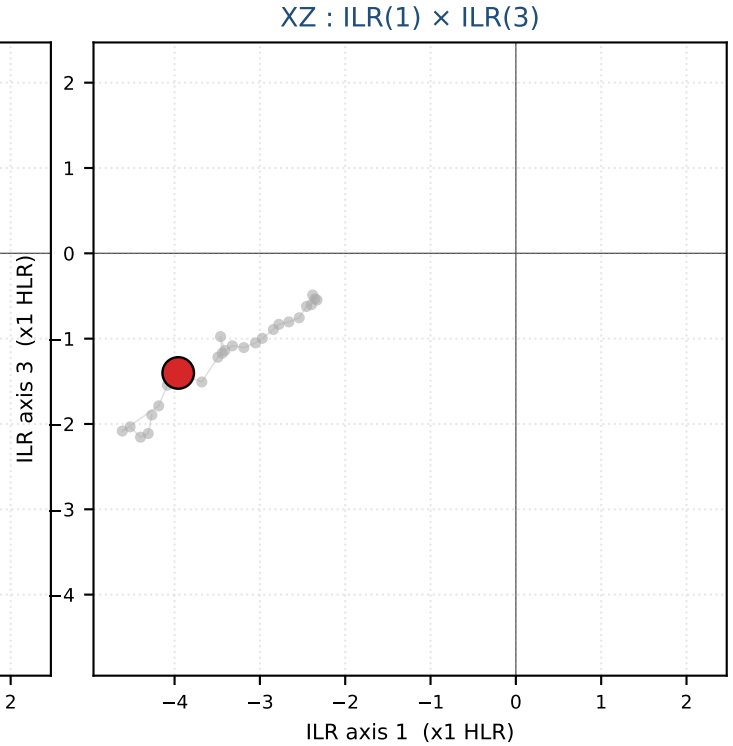
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=7 | 2007 | D=8 T=26 | level 1 first-principles

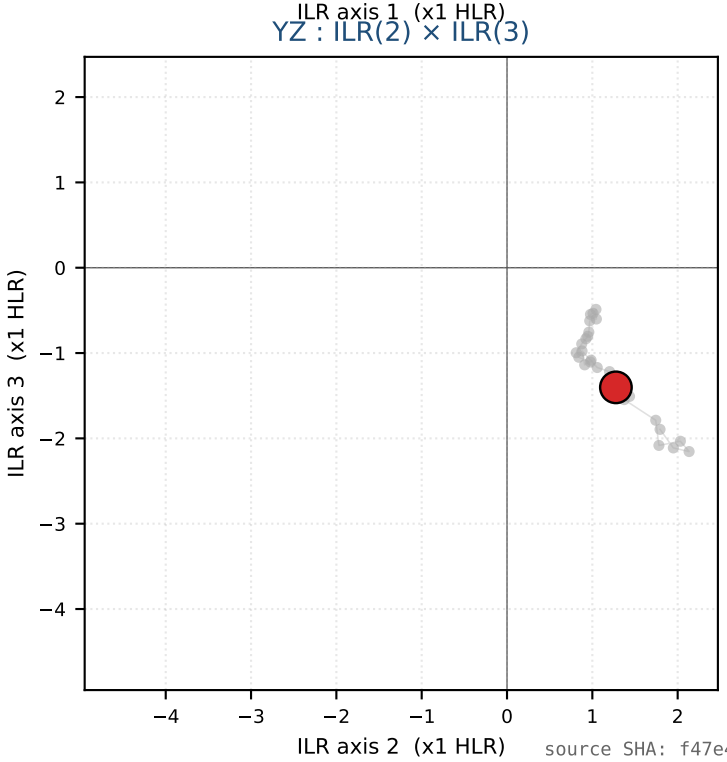
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

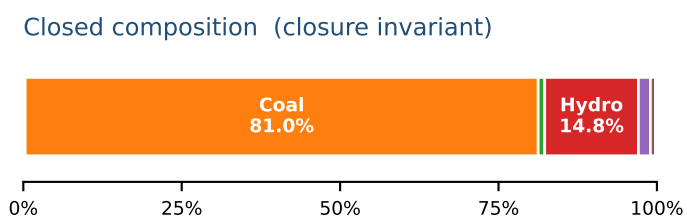




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0030	-1.067
Coal	0.8096	+4.530
Gas	0.0103	+0.169
Hydro	0.1479	+2.830
Nuclear	0.0189	+0.774
Other Fossil	0.0086	-0.020
Solar	0.0000	-5.562
Wind	0.0017	-1.654





closure: Σ = 1.000000 OK

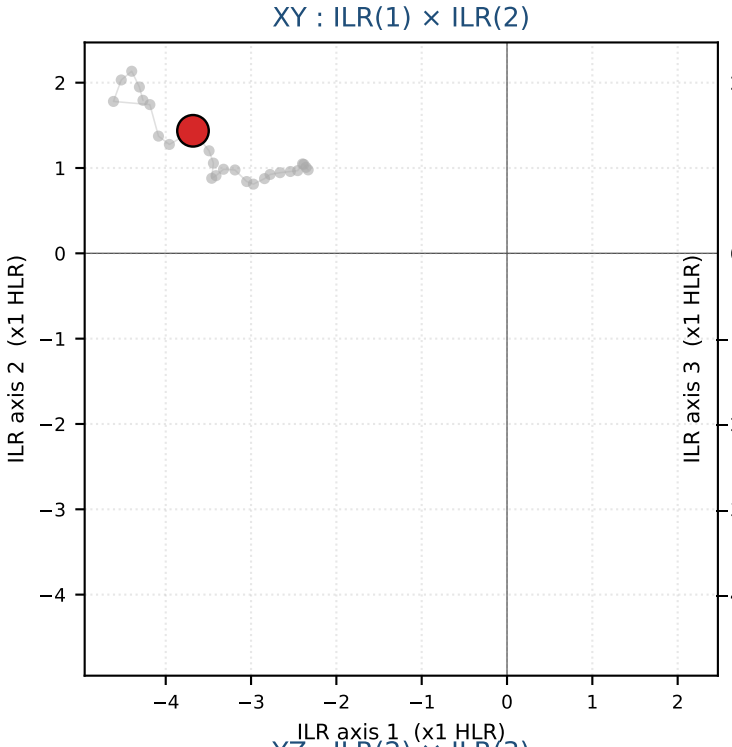
display factor: x1 (raw HLR x 1)

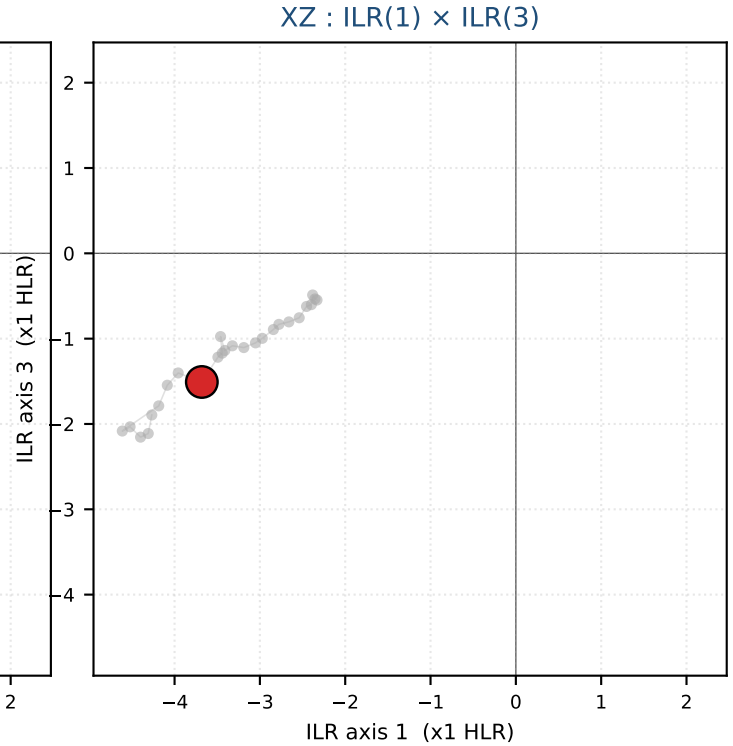
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=8 | 2008 | D=8 T=26 | level 1 first-principles

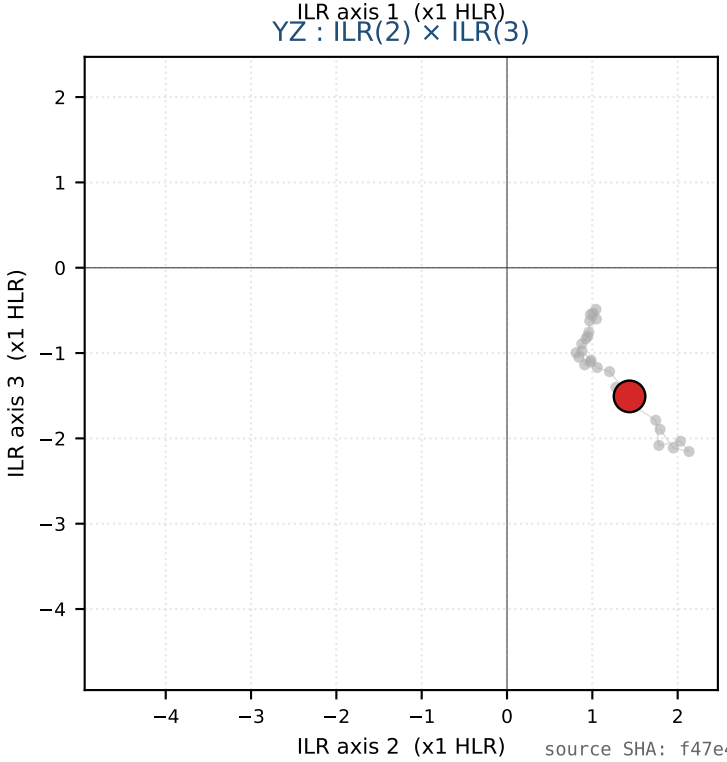
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



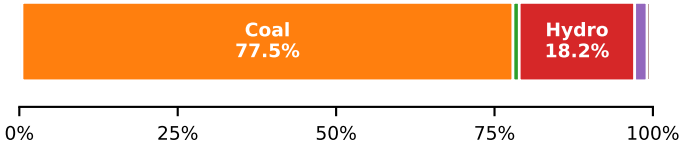


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0043	-0.855
Coal	0.7749	+4.349
Gas	0.0099	-0.012
Hydro	0.1822	+2.902
Nuclear	0.0196	+0.670
Other Fossil	0.0054	-0.620
Solar	0.0000	-5.452
Wind	0.0037	-0.982



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

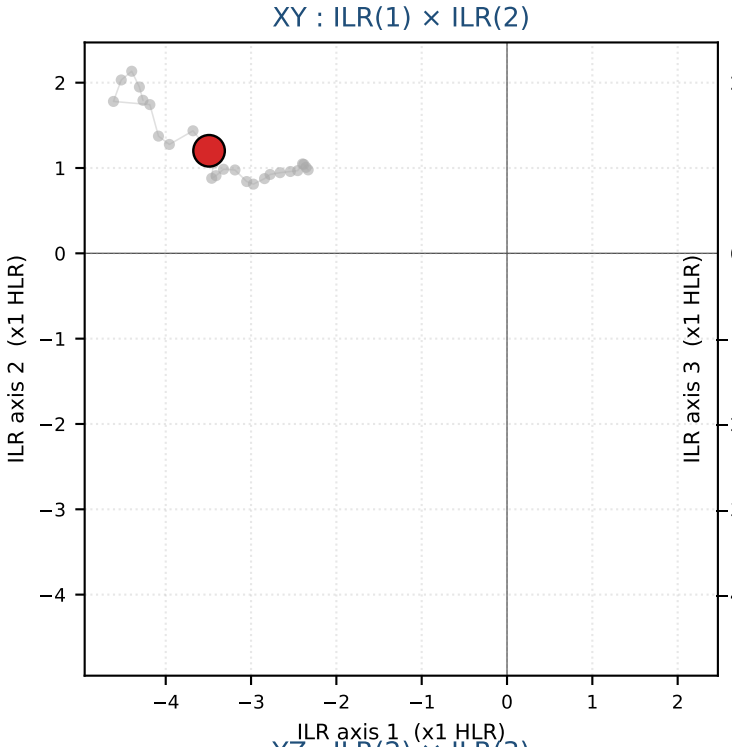
display factor: x1 (raw HLR x 1)

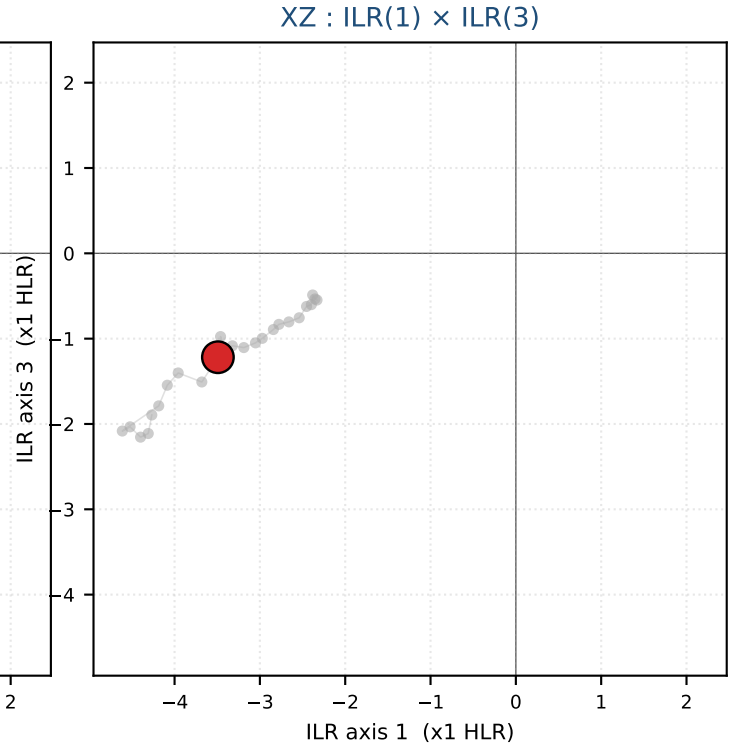
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=9 | 2009 | D=8 T=26 | level 1 first-principles

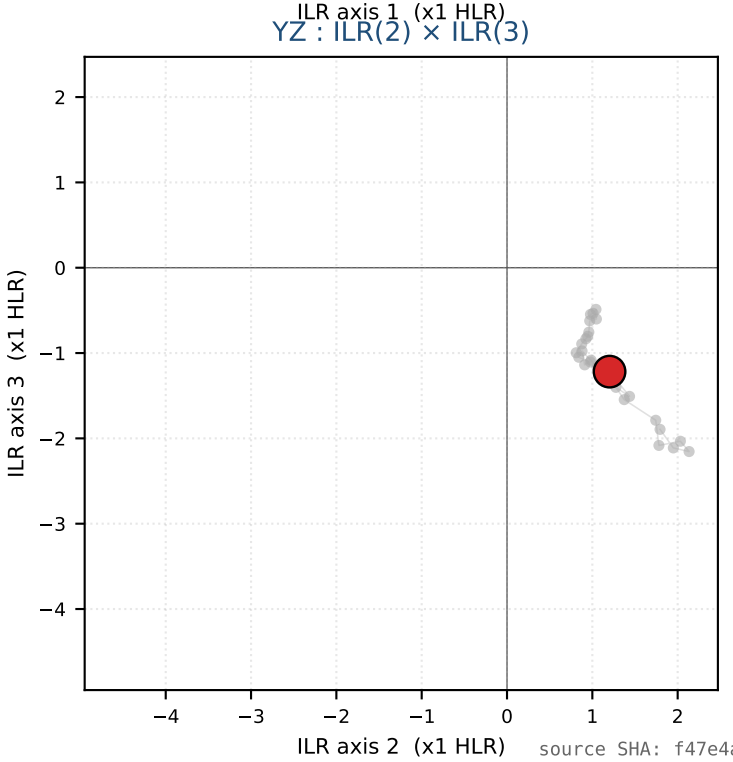
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

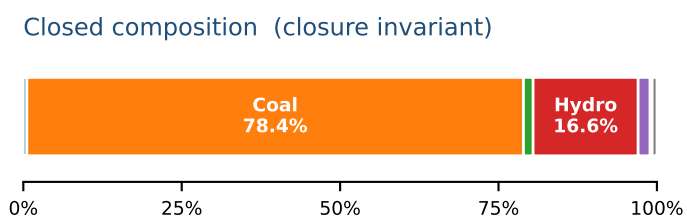




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0056	-0.745
Coal	0.7838	+4.194
Gas	0.0152	+0.253
Hydro	0.1657	+2.640
Nuclear	0.0189	+0.467
Other Fossil	0.0033	-1.288
Solar	0.0001	-5.056
Wind	0.0074	-0.465





closure: Σ = 1.000000 OK

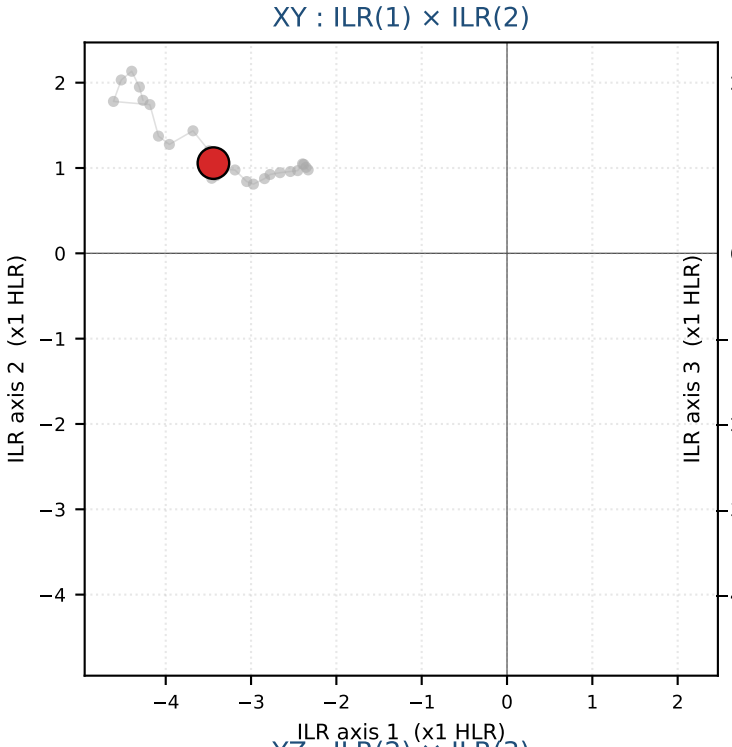
display factor: x1 (raw HLR x 1)

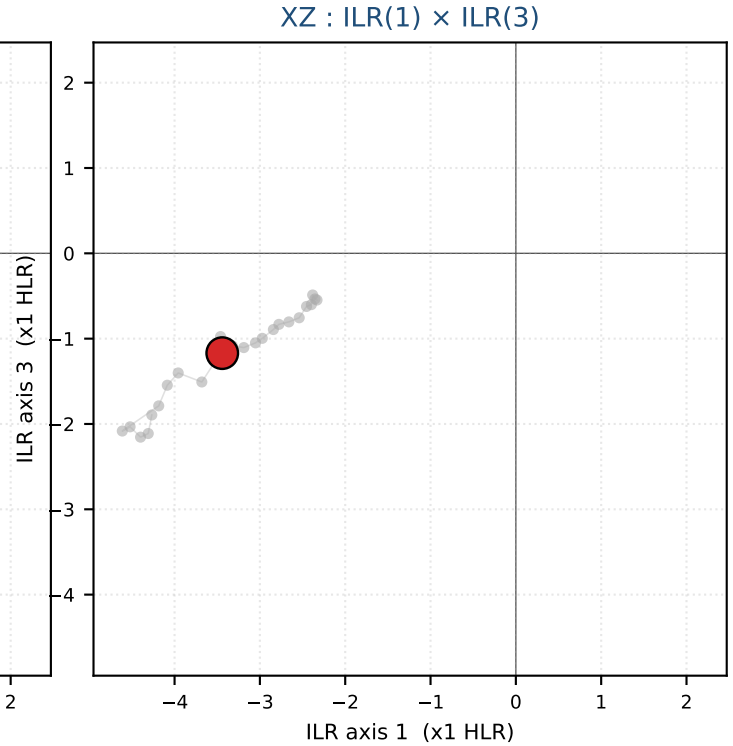
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=10 | 2010 | D=8 T=26 | level 1 first-principles

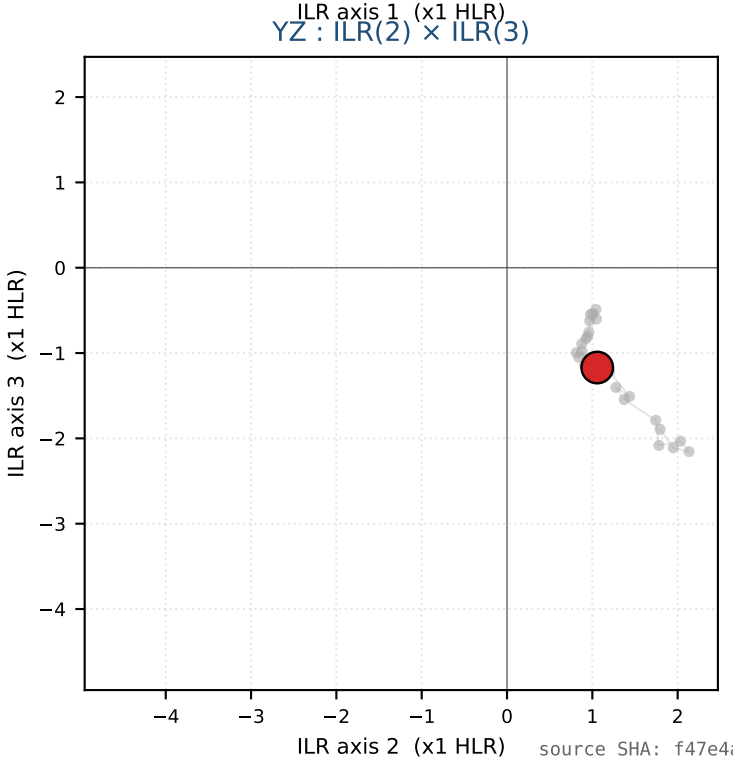
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

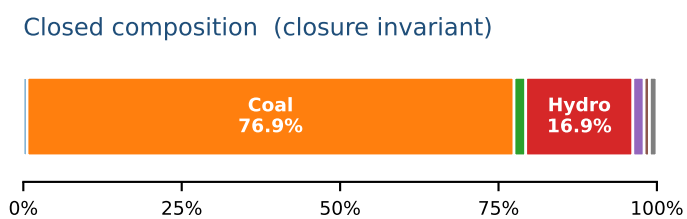




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0059	-0.988
Coal	0.7686	+3.879
Gas	0.0185	+0.151
Hydro	0.1691	+2.365
Nuclear	0.0178	+0.111
Other Fossil	0.0083	-0.656
Solar	0.0002	-4.559
Wind	0.0117	-0.303





closure: Σ = 1.000000 OK

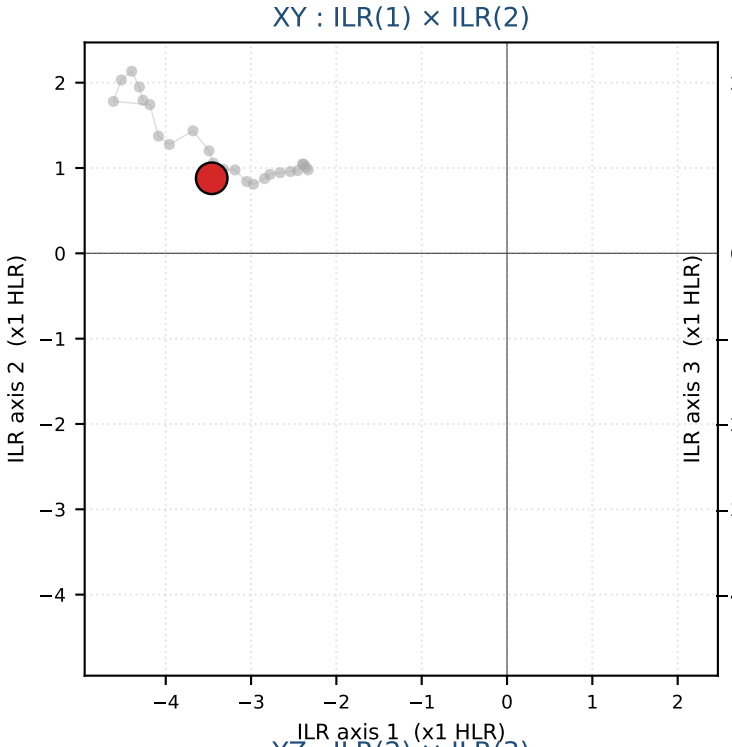
display factor: x1 (raw HLR x 1)

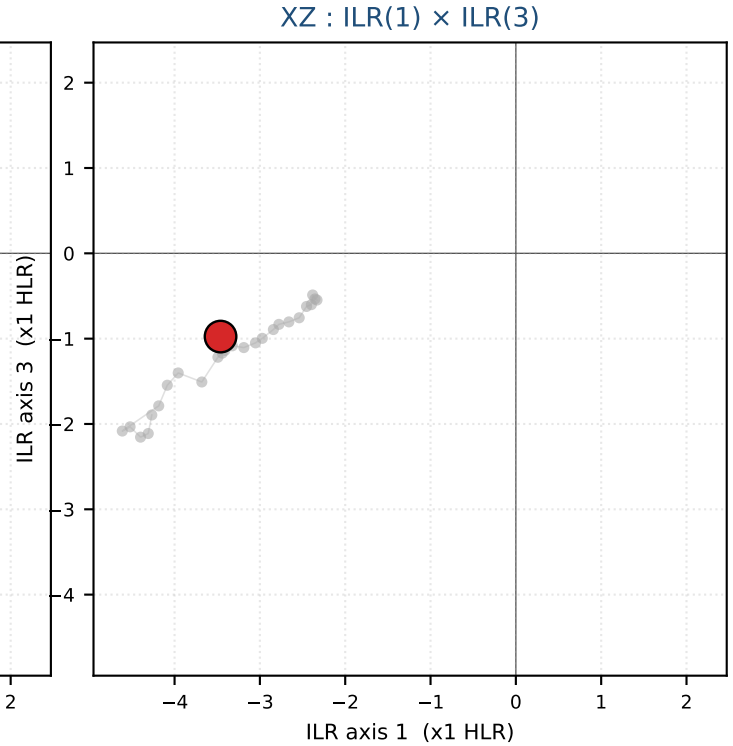
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=11 | 2011 | D=8 T=26 | level 1 first-principles

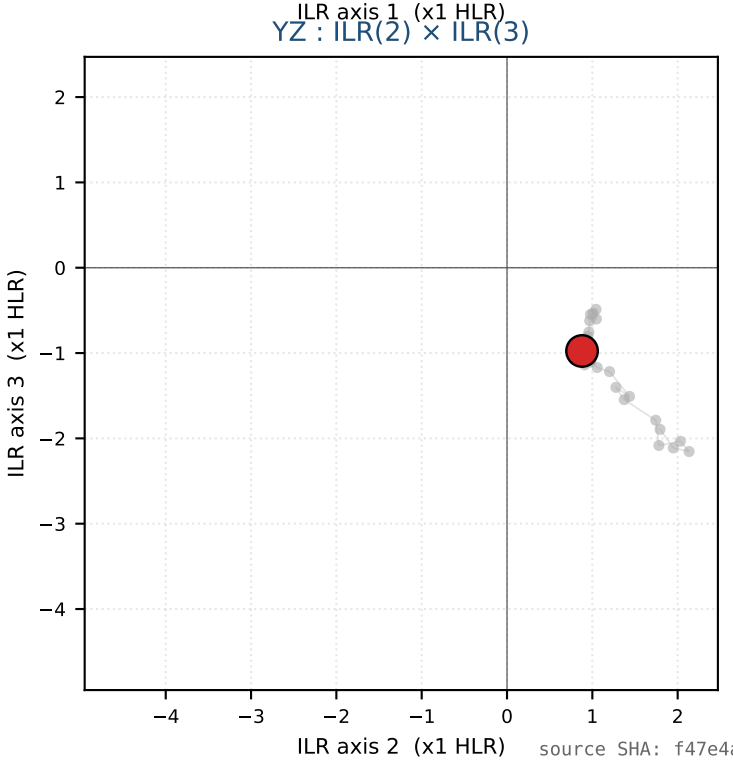
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

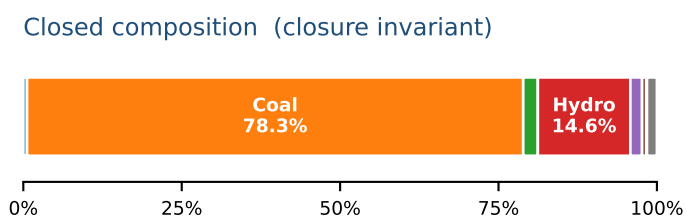




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0059	-1.182
Coal	0.7831	+3.712
Gas	0.0231	+0.188
Hydro	0.1460	+2.033
Nuclear	0.0185	-0.033
Other Fossil	0.0072	-0.981
Solar	0.0006	-3.542
Wind	0.0157	-0.196





closure: Σ = 1.000000 OK

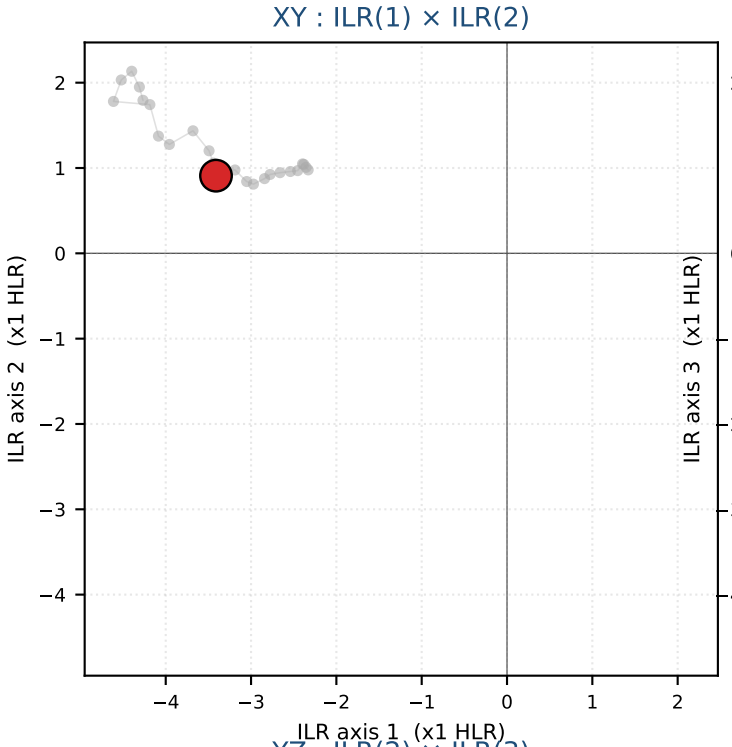
display factor: x1 (raw HLR x 1)

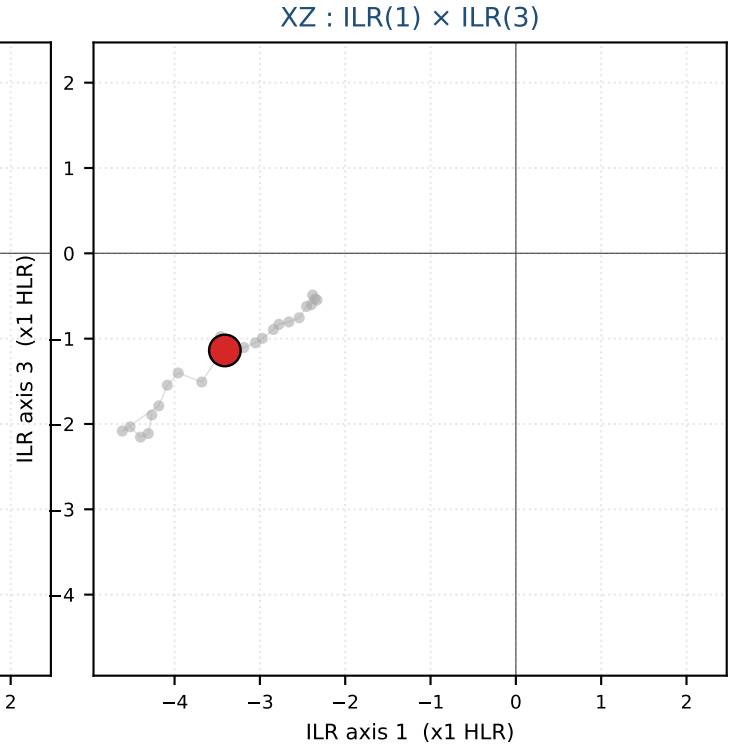
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=12 | 2012 | D=8 T=26 | level 1 first-principles

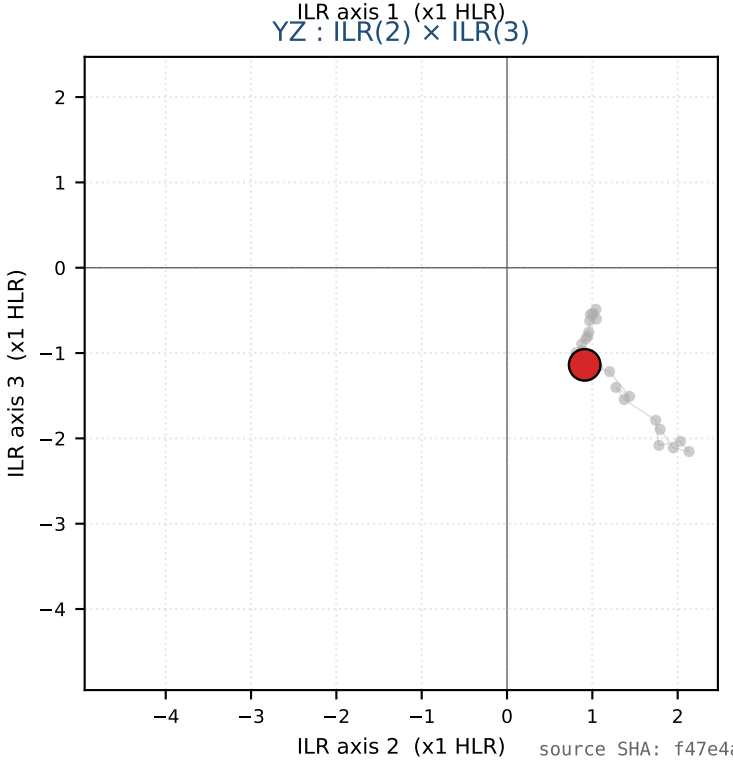
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

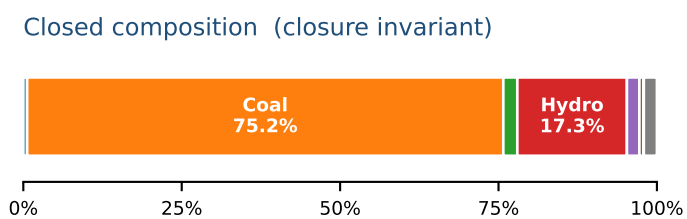




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0060	-1.224
Coal	0.7515	+3.599
Gas	0.0221	+0.073
Hydro	0.1730	+2.130
Nuclear	0.0197	-0.042
Other Fossil	0.0063	-1.189
Solar	0.0007	-3.352
Wind	0.0207	+0.005





closure: Σ = 1.000000 OK

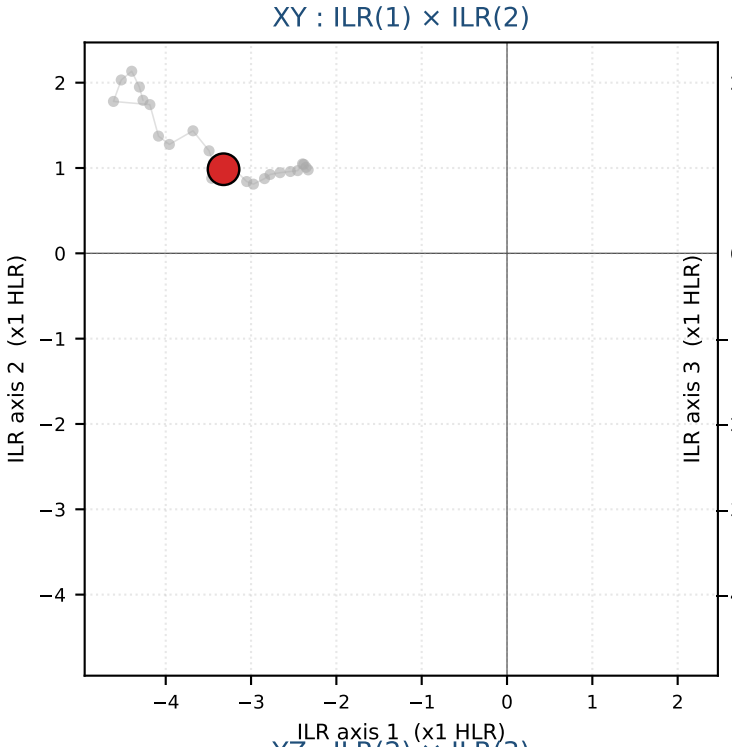
display factor: x1 (raw HLR x 1)

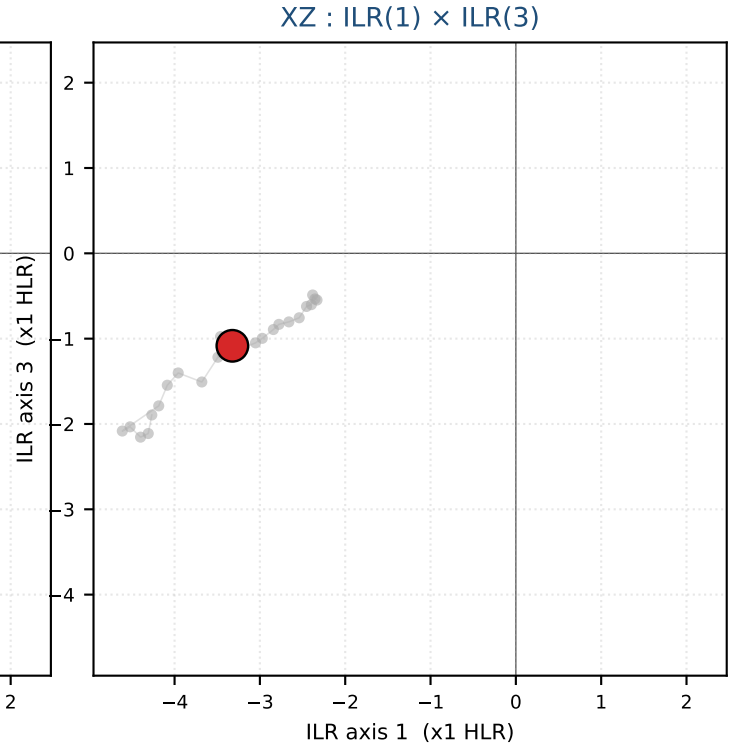
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=13 | 2013 | D=8 T=26 | level 1 first-principles

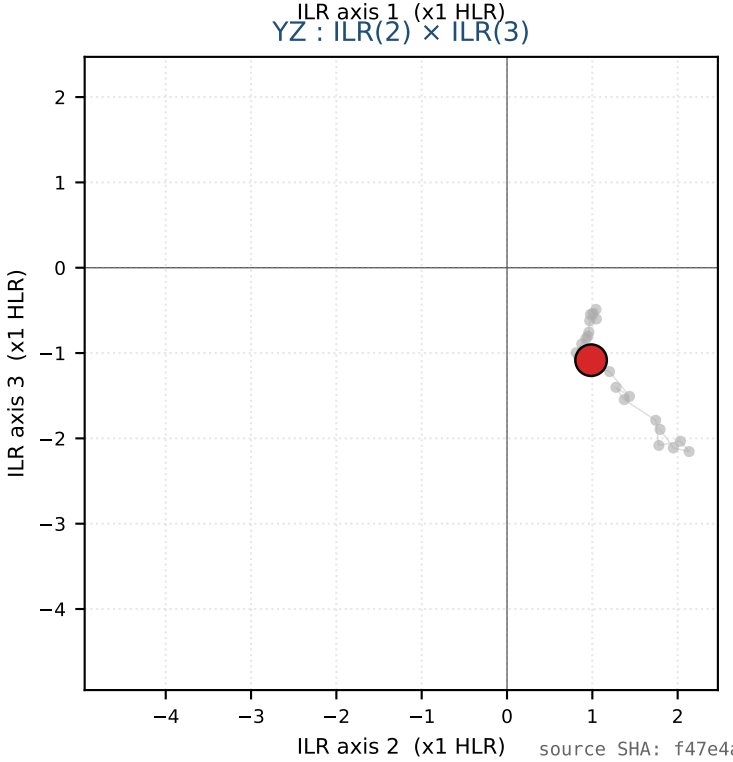
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

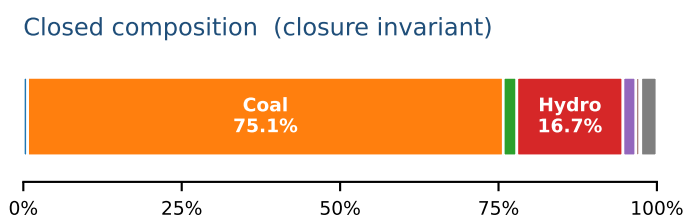




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0068	-1.231
Coal	0.7507	+3.468
Gas	0.0214	-0.088
Hydro	0.1675	+1.968
Nuclear	0.0205	-0.131
Other Fossil	0.0061	-1.349
Solar	0.0015	-2.721
Wind	0.0255	+0.084





closure: Σ = 1.000000 OK

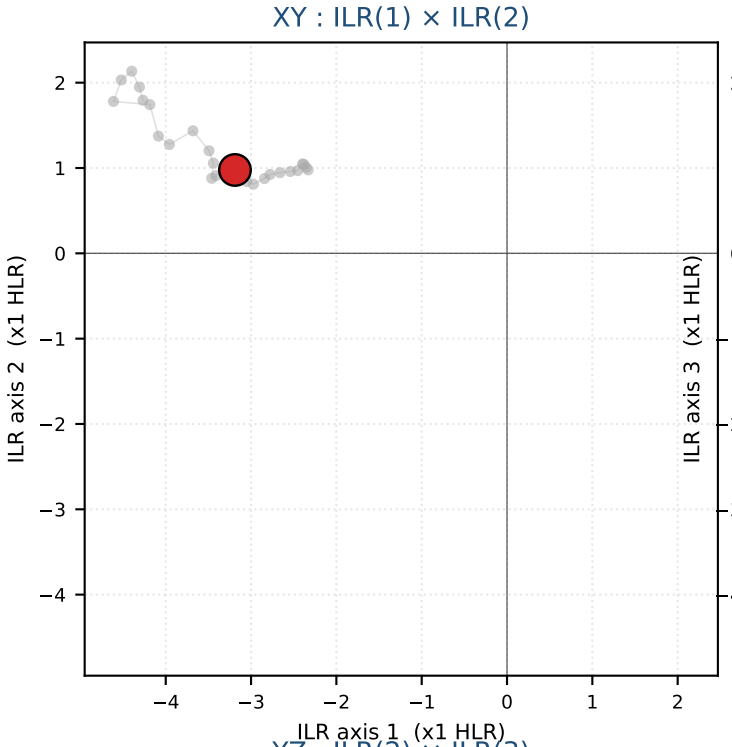
display factor: x1 (raw HLR x 1)

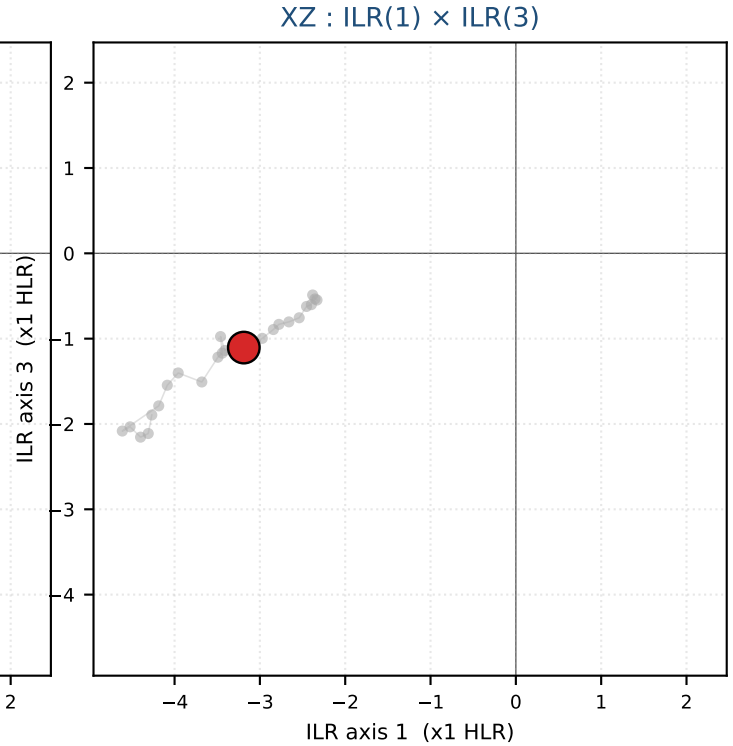
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=14 | 2014 | D=8 T=26 | level 1 first-principles

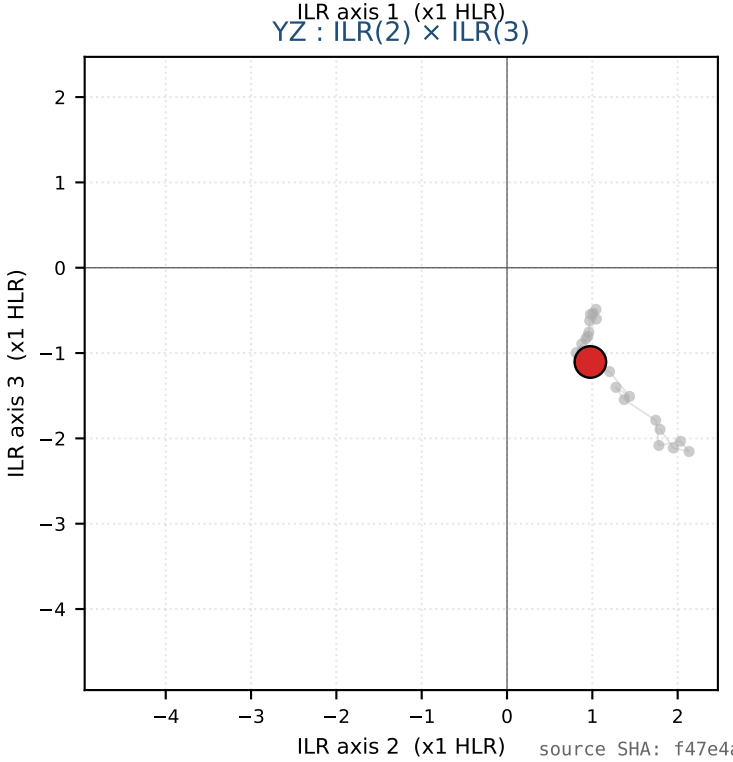
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

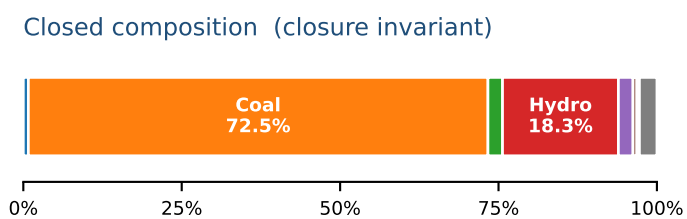




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0080	-1.257
Coal	0.7254	+3.252
Gas	0.0230	-0.199
Hydro	0.1829	+1.874
Nuclear	0.0230	-0.200
Other Fossil	0.0062	-1.518
Solar	0.0041	-1.934
Wind	0.0276	-0.018





closure: Σ = 1.000000 OK

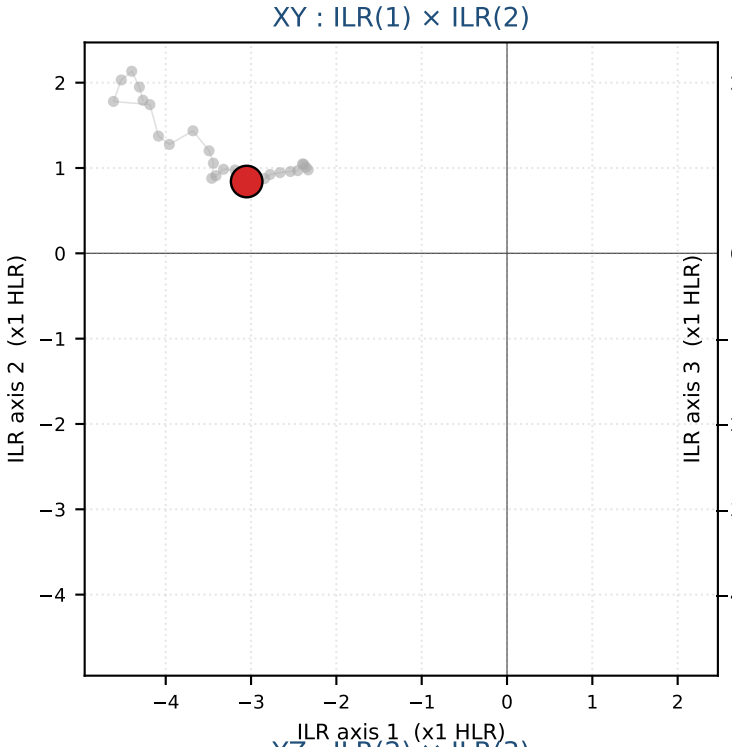
display factor: x1 (raw HLR x 1)

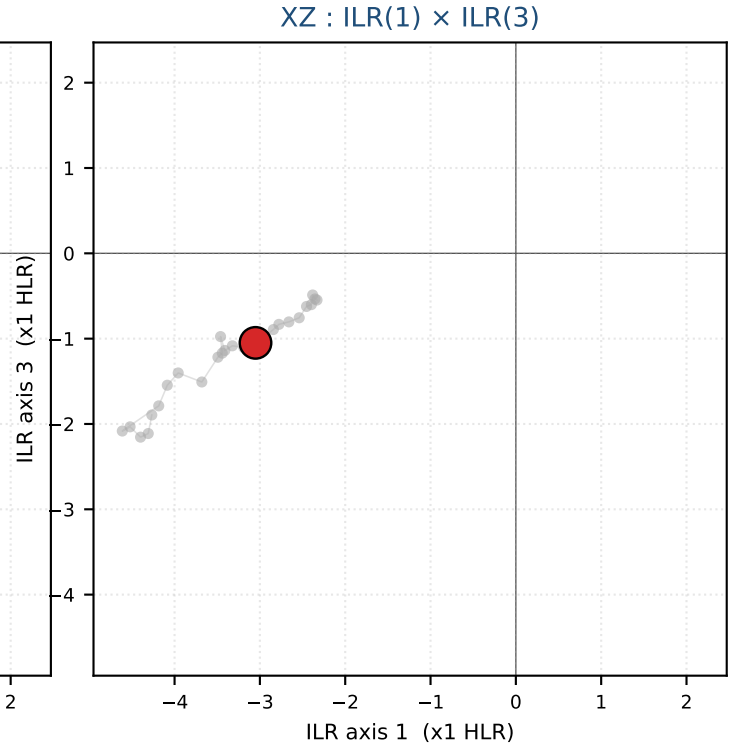
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=15 | 2015 | D=8 T=26 | level 1 first-principles

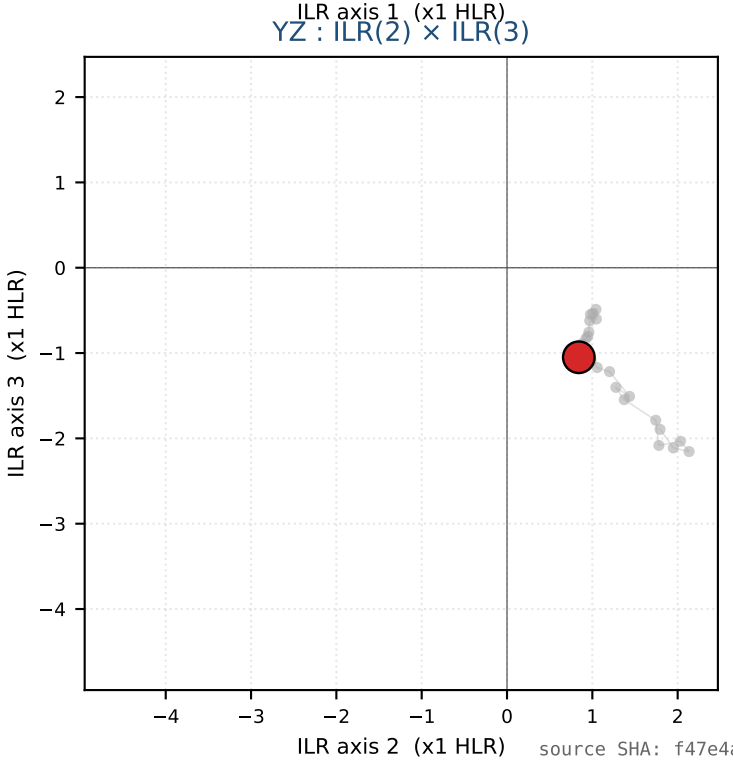
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

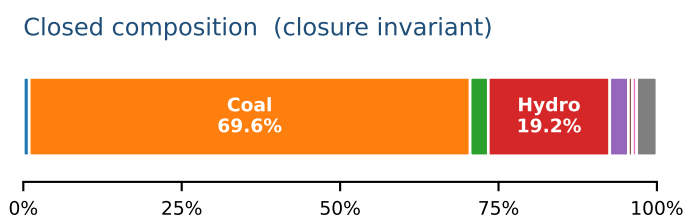




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0093	-1.268
Coal	0.6959	+3.047
Gas	0.0287	-0.141
Hydro	0.1917	+1.758
Nuclear	0.0295	-0.115
Other Fossil	0.0063	-1.662
Solar	0.0068	-1.583
Wind	0.0319	-0.035





closure: Σ = 1.000000 OK

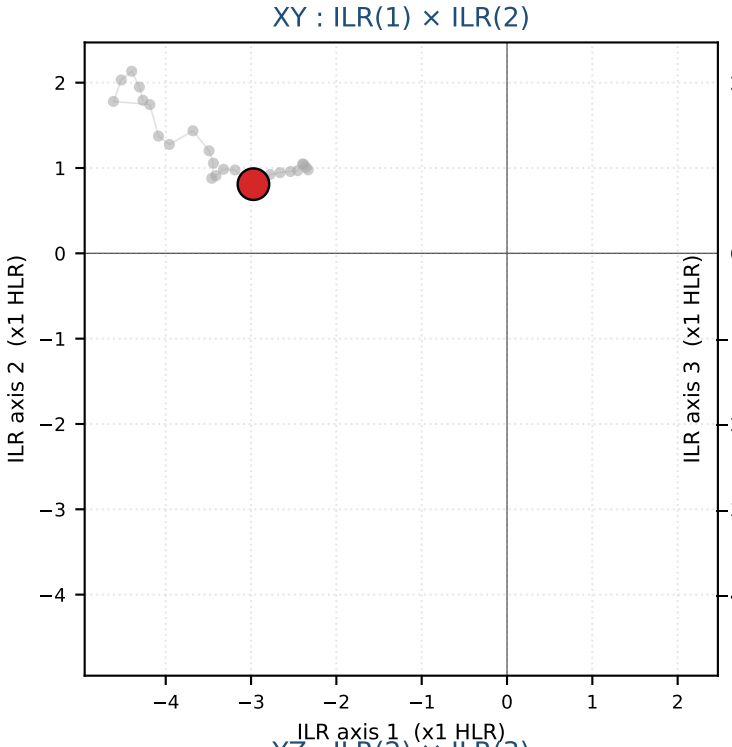
display factor: x1 (raw HLR x 1)

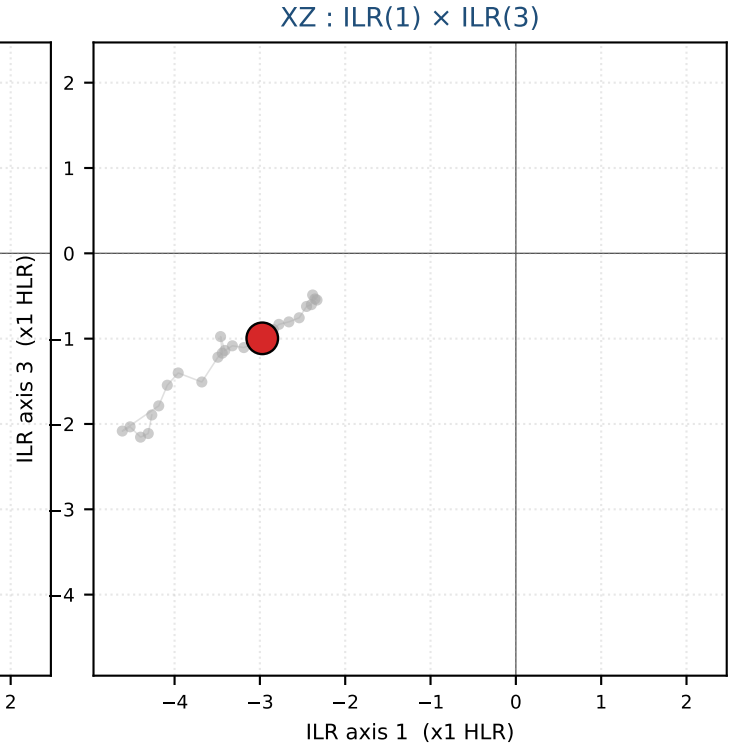
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=16 | 2016 | D=8 T=26 | level 1 first-principles

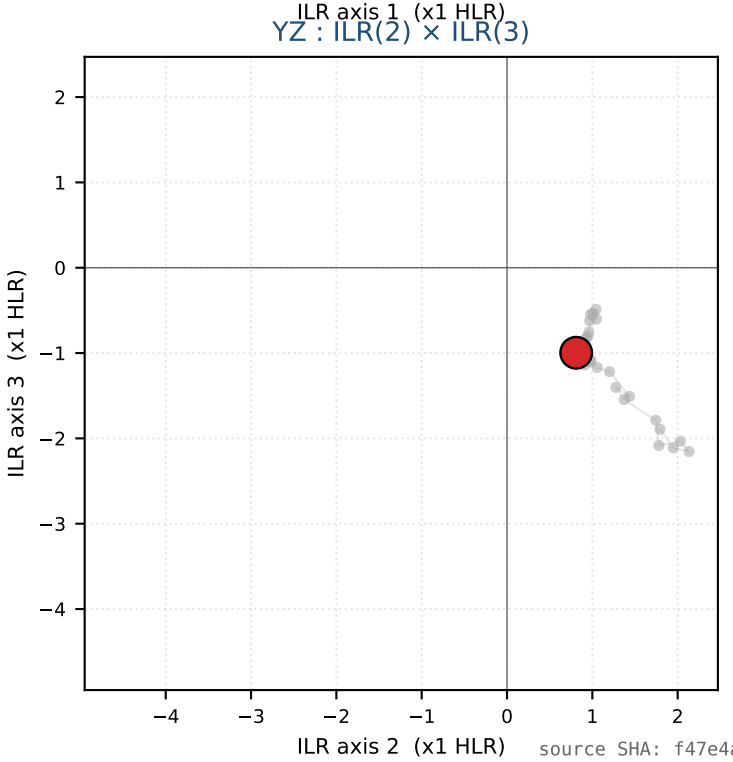
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

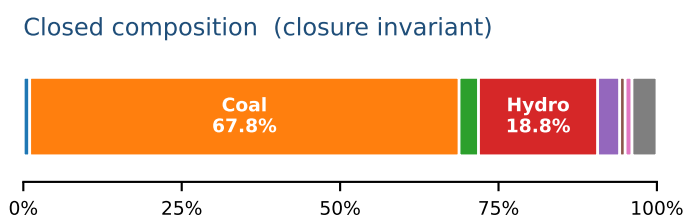




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0101	-1.340
Coal	0.6777	+2.863
Gas	0.0307	-0.232
Hydro	0.1880	+1.581
Nuclear	0.0348	-0.107
Other Fossil	0.0086	-1.508
Solar	0.0108	-1.272
Wind	0.0393	+0.015





closure: Σ = 1.000000 OK

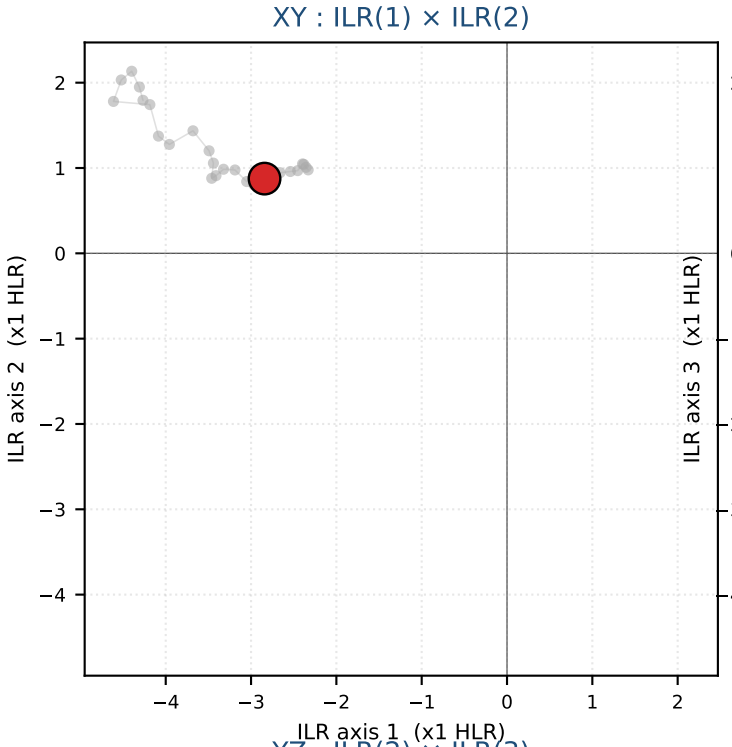
display factor: x1 (raw HLR x 1)

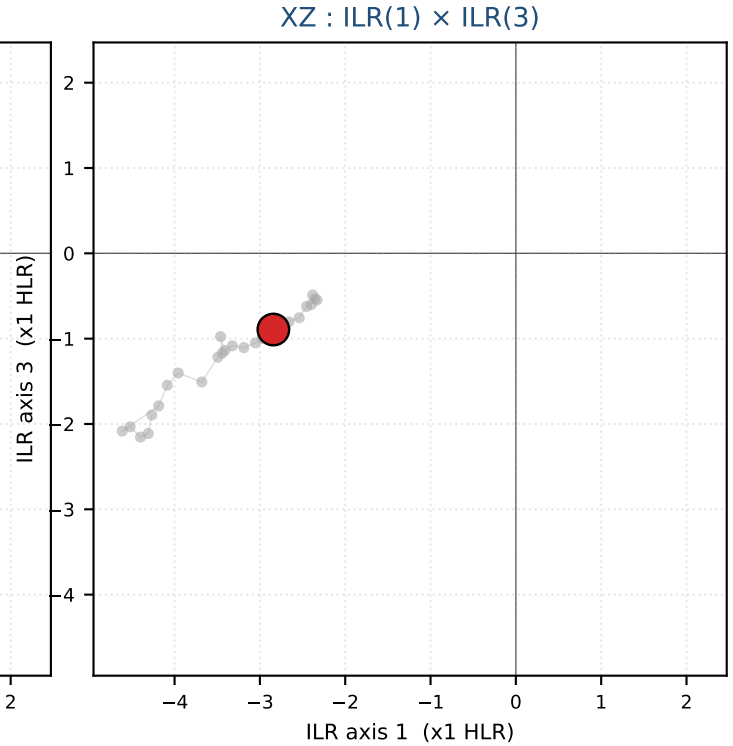
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=17 | 2017 | D=8 T=26 | level 1 first-principles

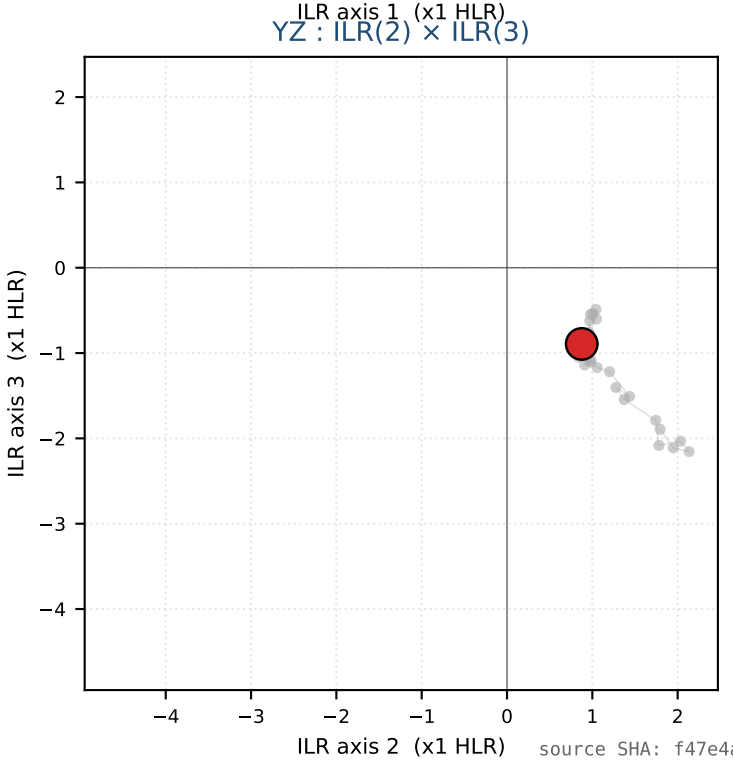
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

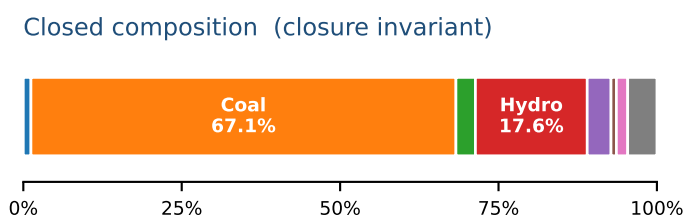




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0121	-1.270
Coal	0.6708	+2.749
Gas	0.0308	-0.333
Hydro	0.1764	+1.414
Nuclear	0.0376	-0.133
Other Fossil	0.0085	-1.621
Solar	0.0178	-0.878
Wind	0.0461	+0.072





closure: Σ = 1.000000 OK

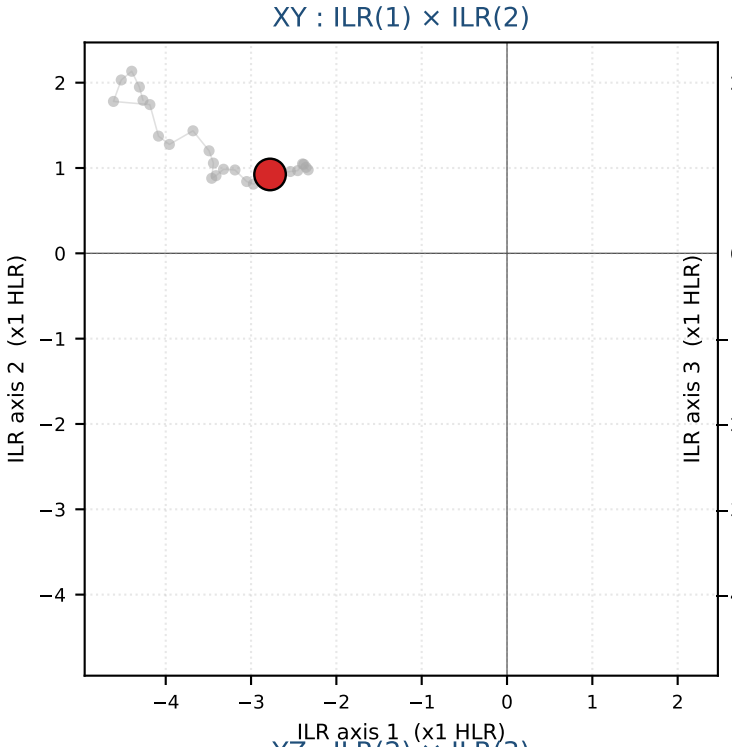
display factor: x1 (raw HLR x 1)

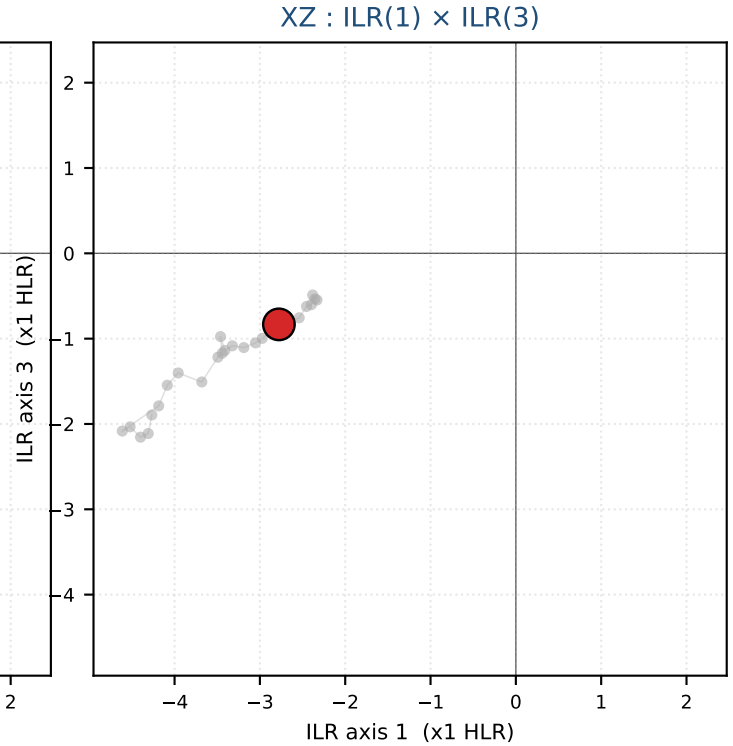
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=18 | 2018 | D=8 T=26 | level 1 first-principles

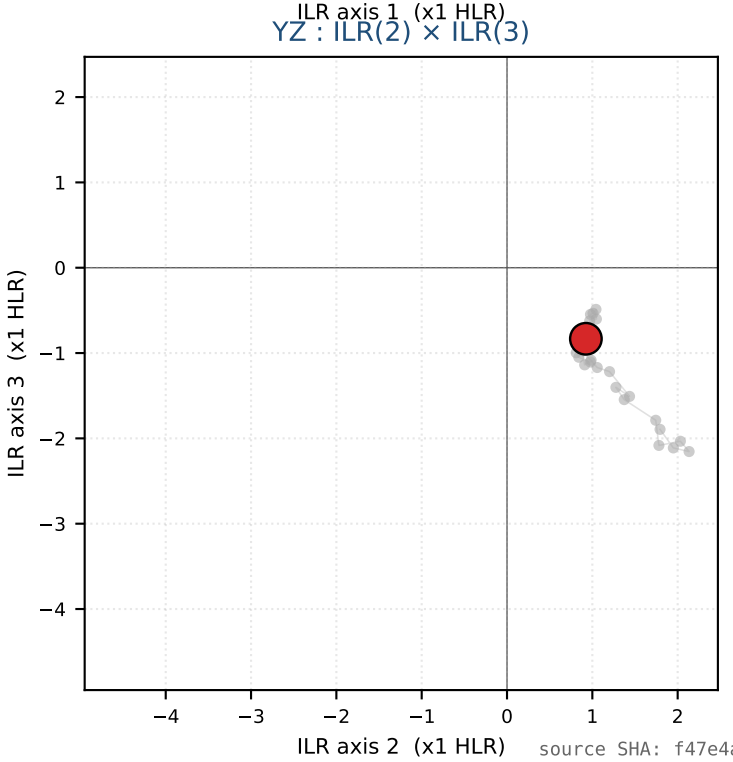
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



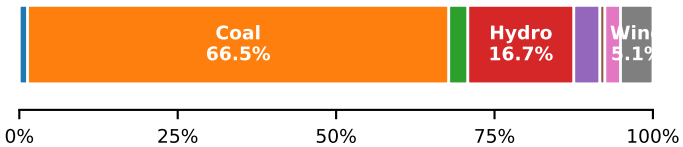


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0131	-1.242
Coal	0.6648	+2.685
Gas	0.0301	-0.411
Hydro	0.1673	+1.306
Nuclear	0.0412	-0.097
Other Fossil	0.0079	-1.751
Solar	0.0247	-0.608
Wind	0.0510	+0.118



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

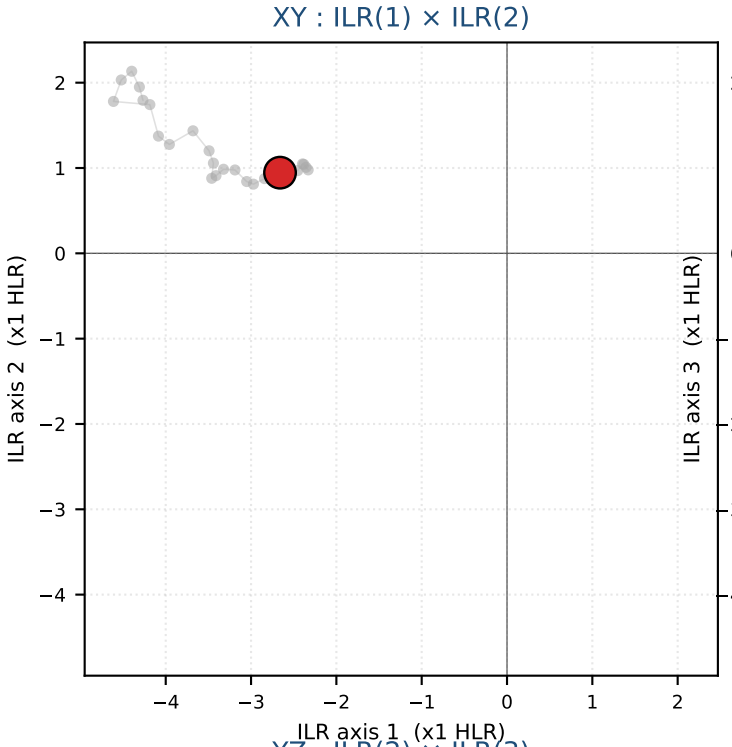
display factor: x1 (raw HLR x 1)

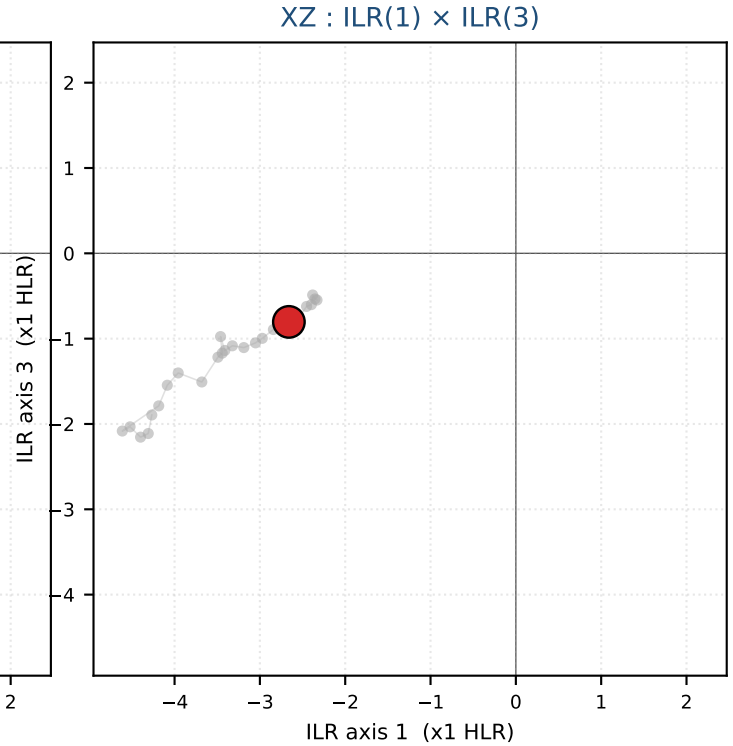
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=19 | 2019 | D=8 T=26 | level 1 first-principles

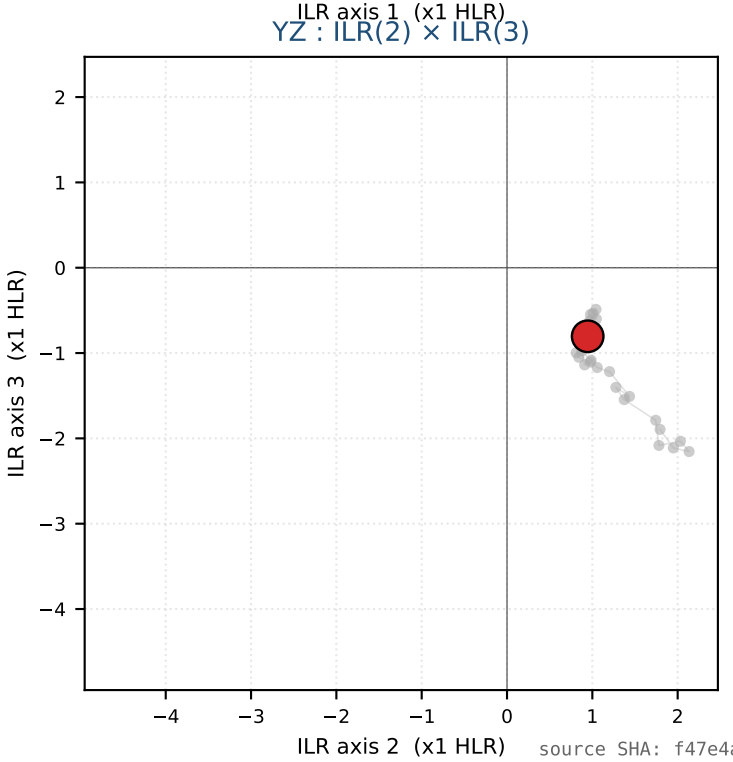
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

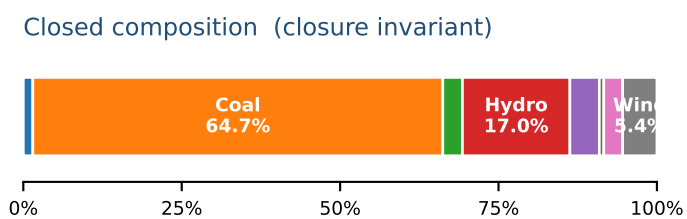




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0150	-1.155
Coal	0.6471	+2.608
Gas	0.0310	-0.431
Hydro	0.1696	+1.269
Nuclear	0.0465	-0.026
Other Fossil	0.0070	-1.920
Solar	0.0299	-0.469
Wind	0.0540	+0.124





closure: Σ = 1.000000 OK

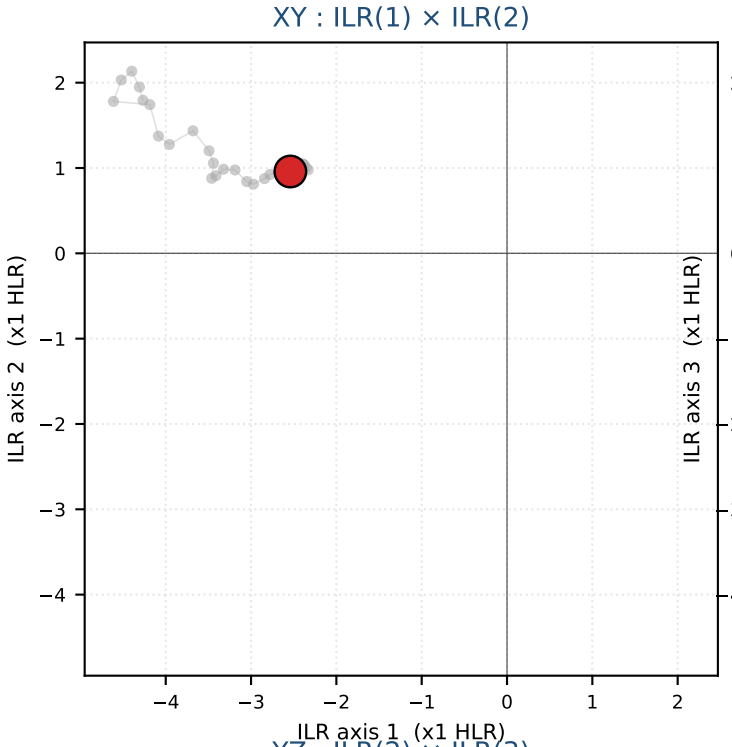
display factor: x1 (raw HLR x 1)

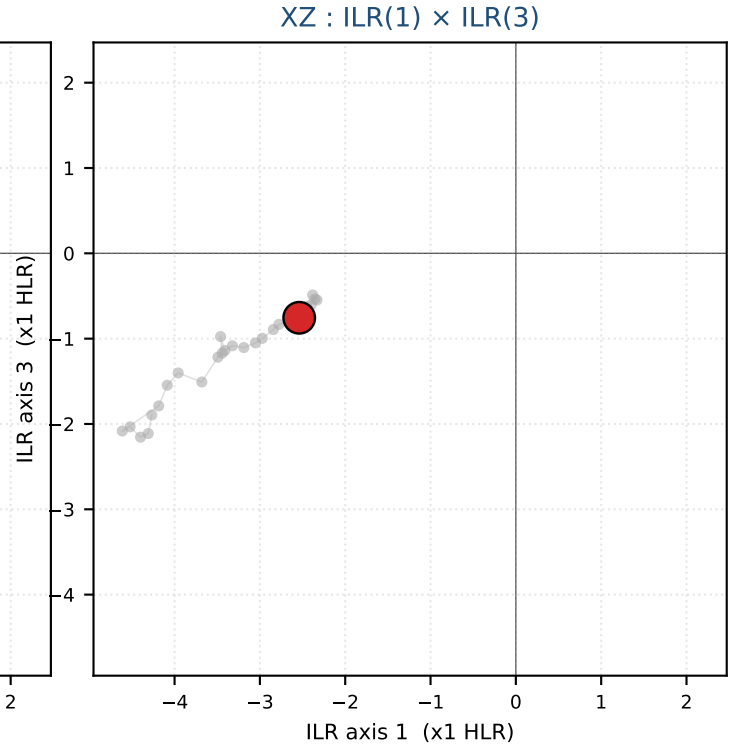
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=20 | 2020 | D=8 T=26 | level 1 first-principles

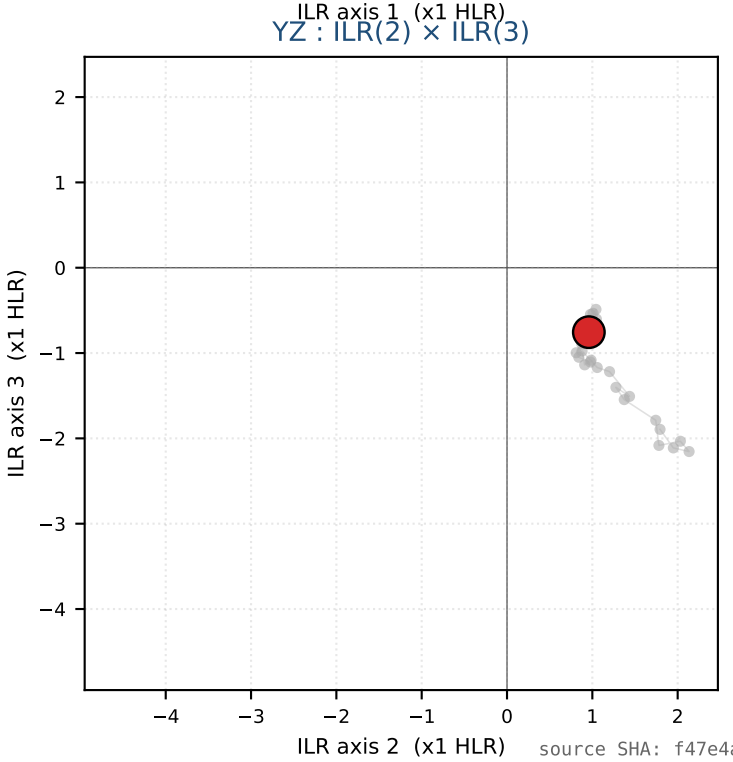
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



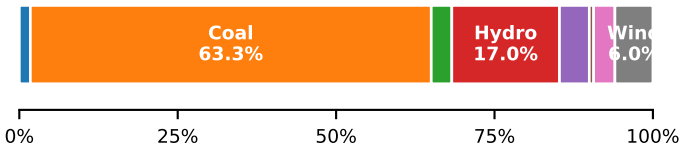


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0174	-1.055
Coal	0.6327	+2.536
Gas	0.0325	-0.434
Hydro	0.1699	+1.221
Nuclear	0.0471	-0.062
Other Fossil	0.0069	-1.986
Solar	0.0336	-0.400
Wind	0.0600	+0.180



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

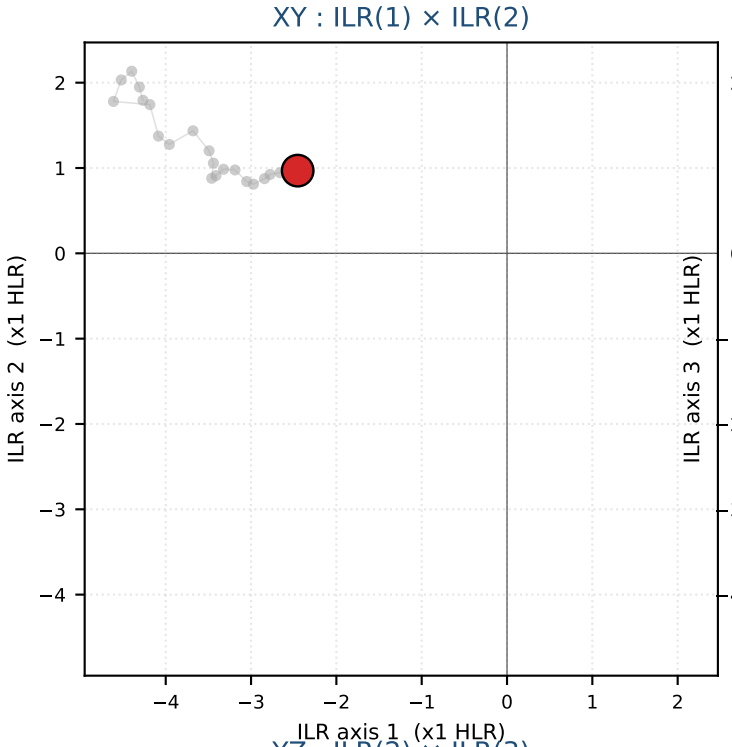
display factor: x1 (raw HLR x 1)

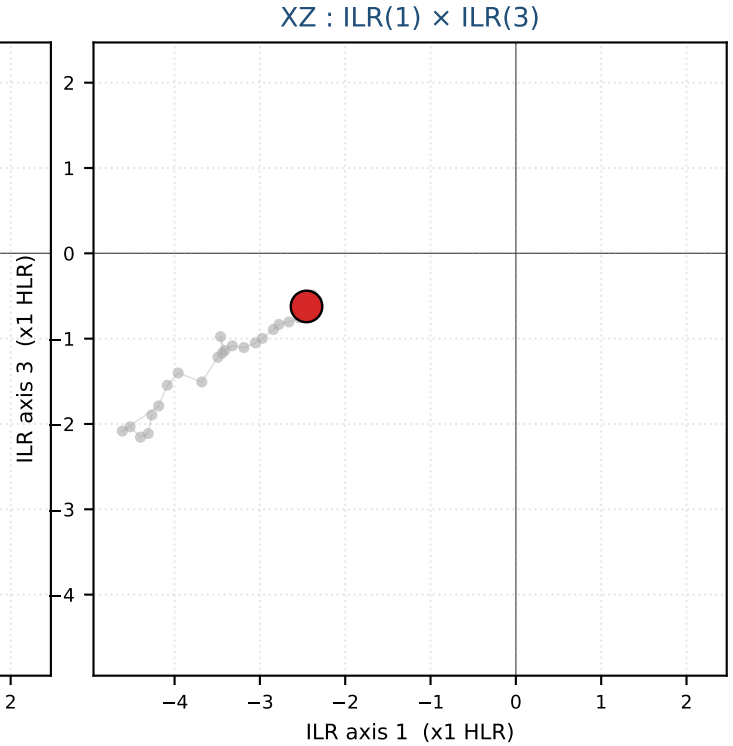
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=21 | 2021 | D=8 T=26 | level 1 first-principles

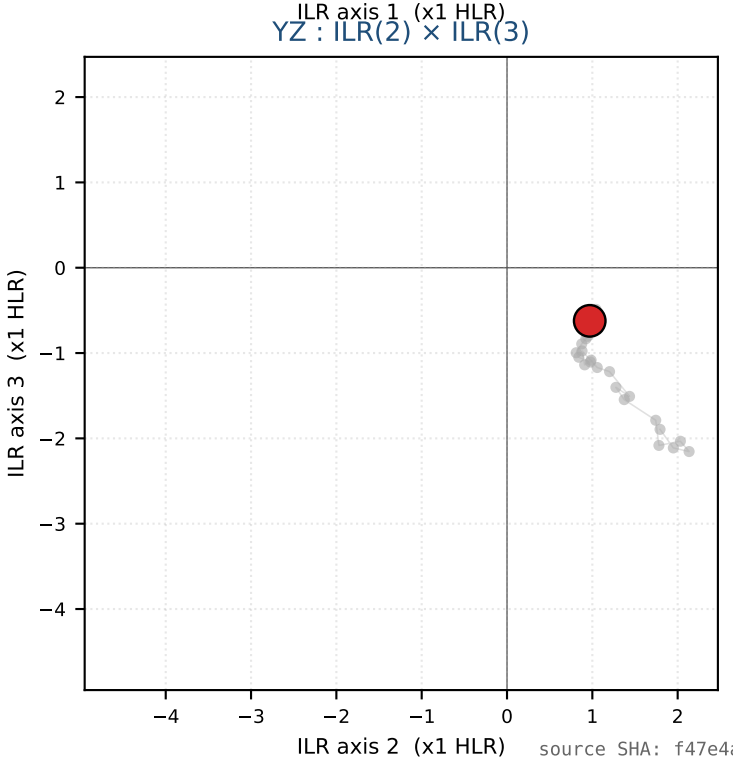
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

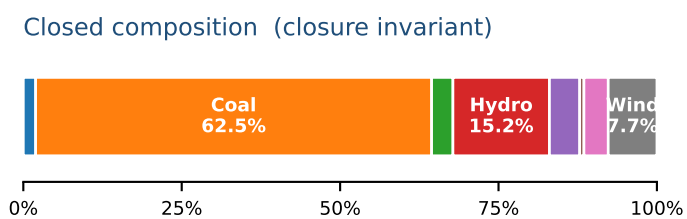




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0195	-0.995
Coal	0.6250	+2.475
Gas	0.0336	-0.447
Hydro	0.1523	+1.063
Nuclear	0.0477	-0.097
Other Fossil	0.0067	-2.063
Solar	0.0383	-0.317
Wind	0.0768	+0.379





closure: Σ = 1.000000 OK

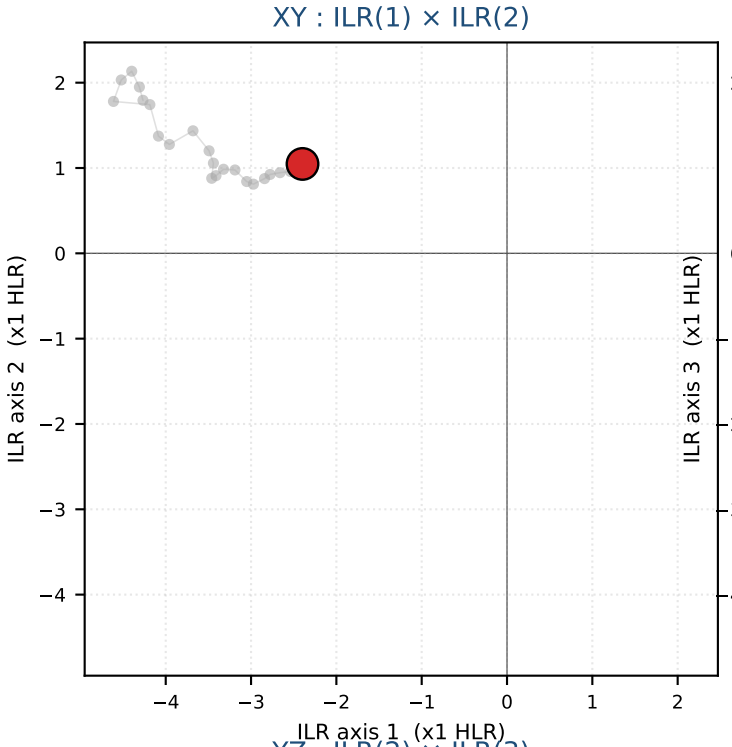
display factor: x1 (raw HLR x 1)

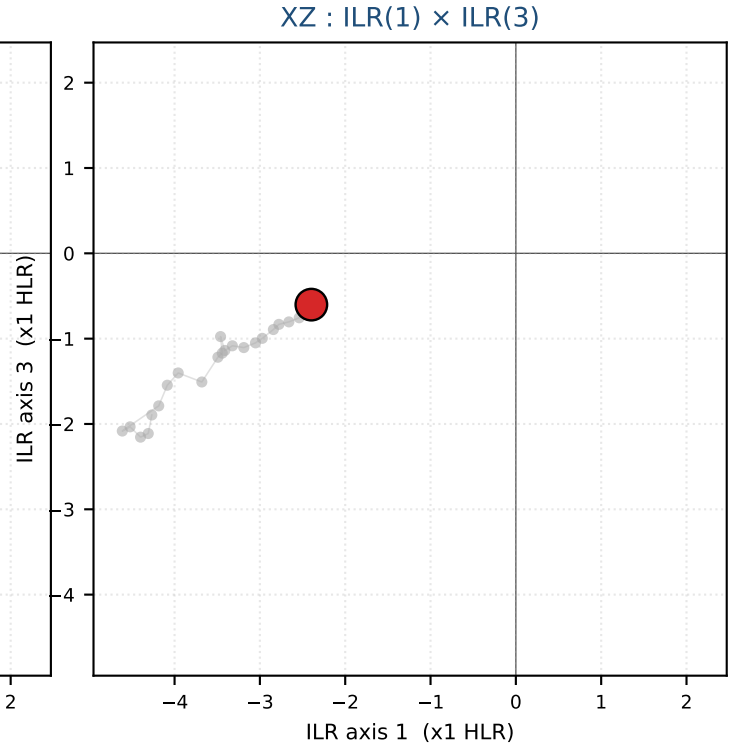
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=22 | 2022 | D=8 T=26 | level 1 first-principles

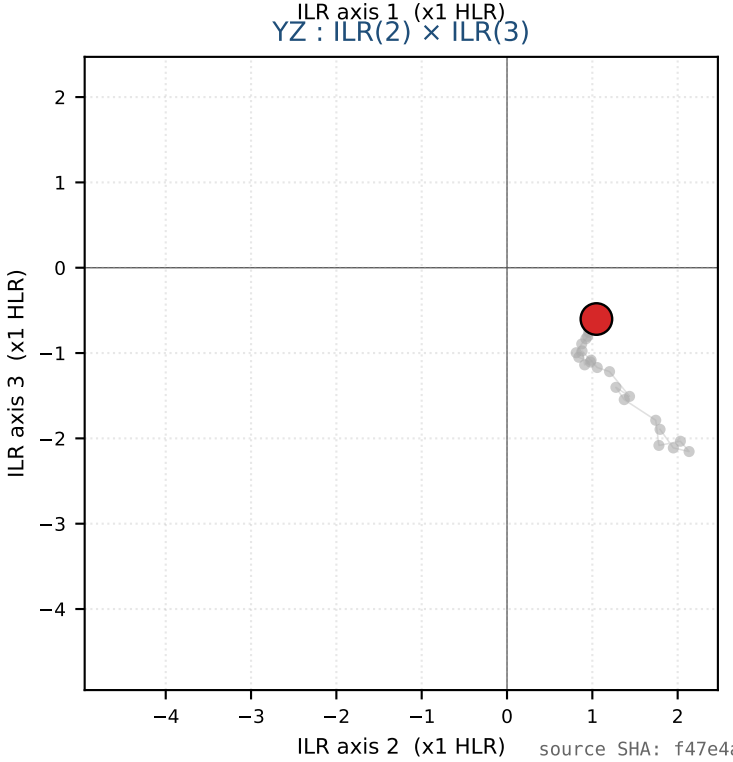
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

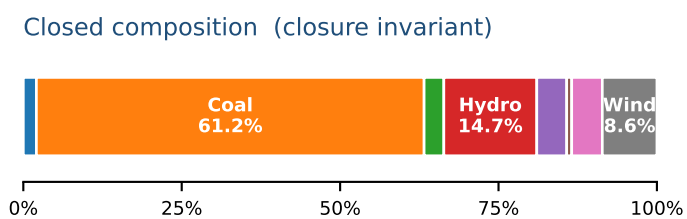




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0206	-0.989
Coal	0.6119	+2.401
Gas	0.0311	-0.577
Hydro	0.1467	+0.973
Nuclear	0.0472	-0.161
Other Fossil	0.0079	-1.949
Solar	0.0483	-0.139
Wind	0.0862	+0.441





closure: Σ = 1.000000 OK

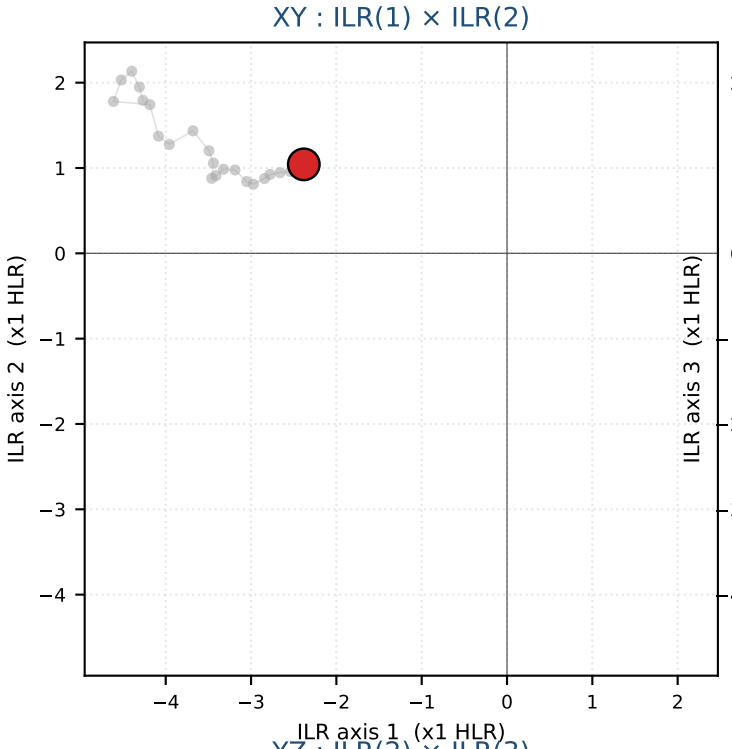
display factor: x1 (raw HLR x 1)

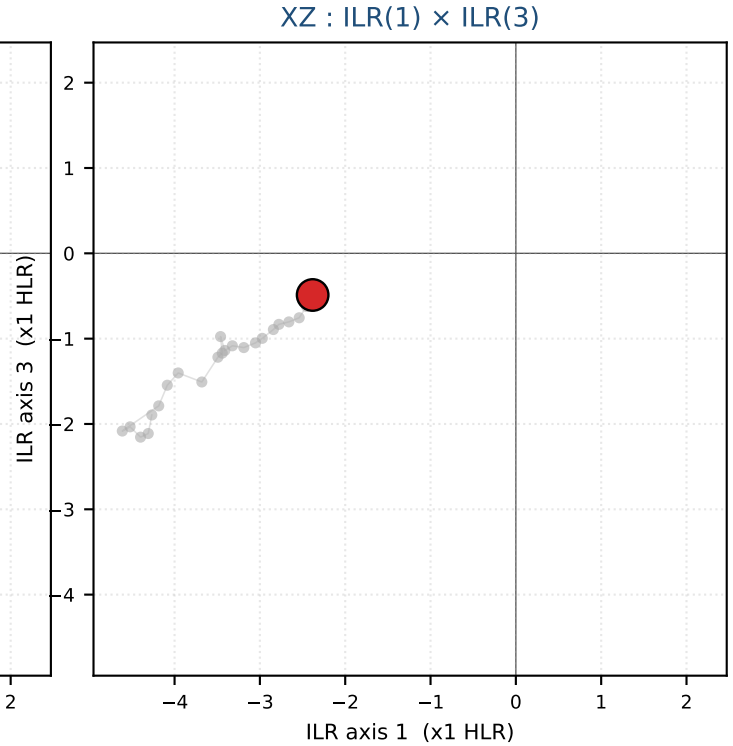
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=23 | 2023 | D=8 T=26 | level 1 first-principles

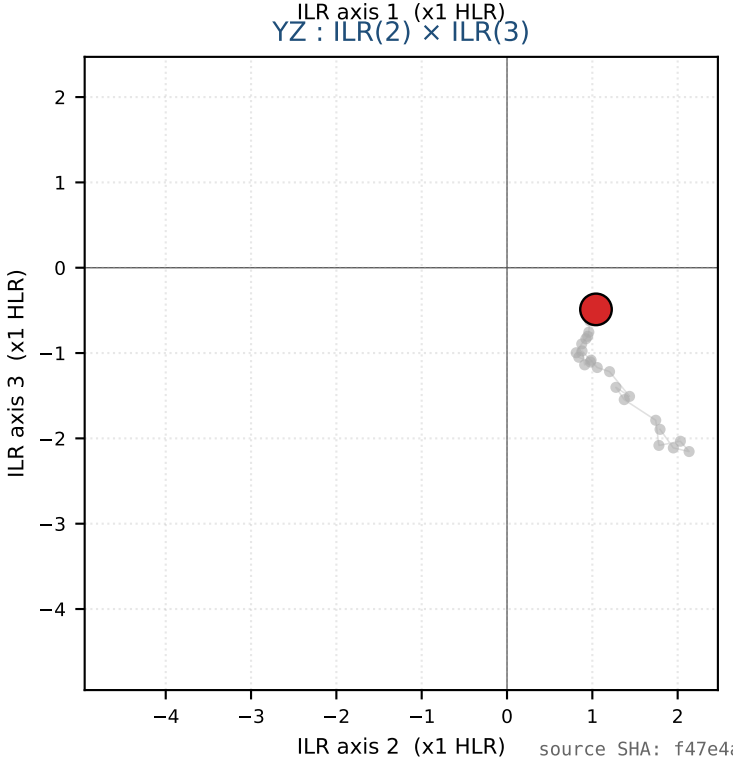
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

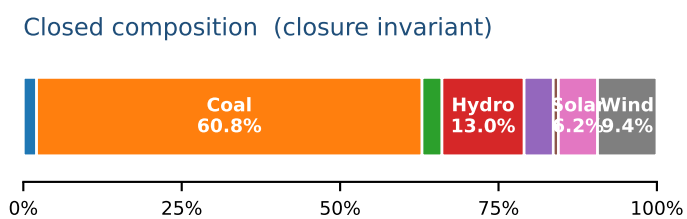




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0210	-1.002
Coal	0.6083	+2.366
Gas	0.0315	-0.595
Hydro	0.1296	+0.820
Nuclear	0.0460	-0.216
Other Fossil	0.0081	-1.948
Solar	0.0618	+0.079
Wind	0.0937	+0.495





closure: Σ = 1.000000 OK

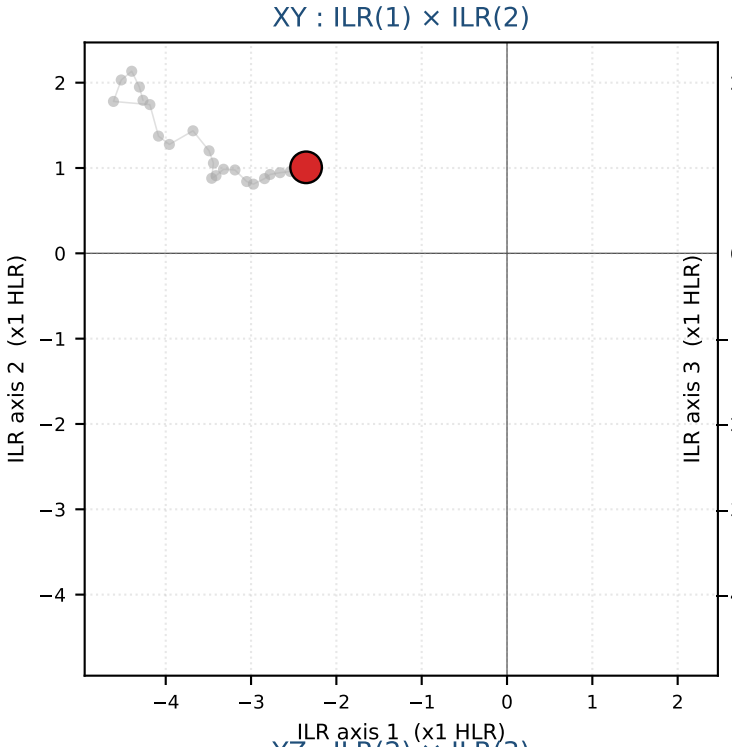
display factor: x1 (raw HLR x 1)

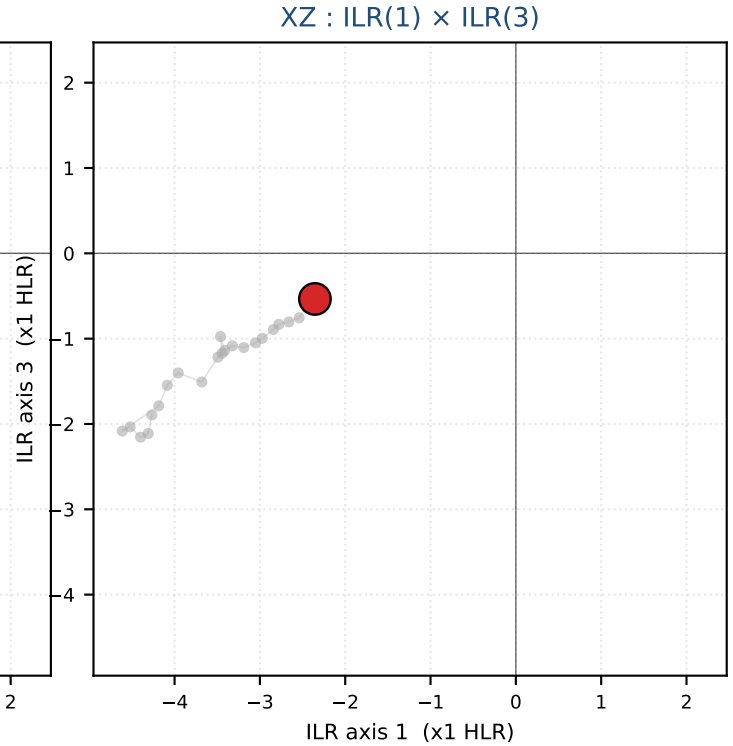
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=24 | 2024 | D=8 T=26 | level 1 first-principles

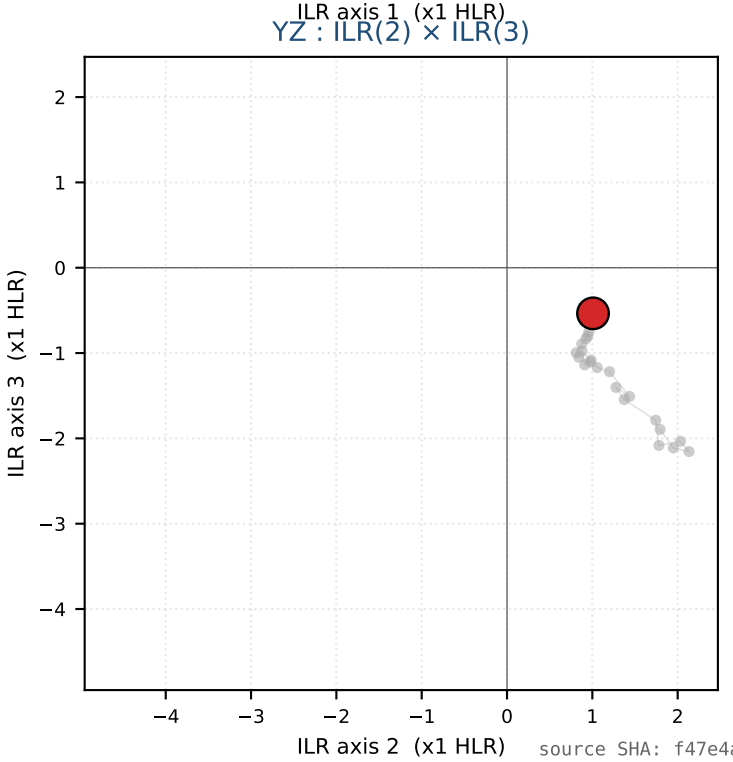
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4



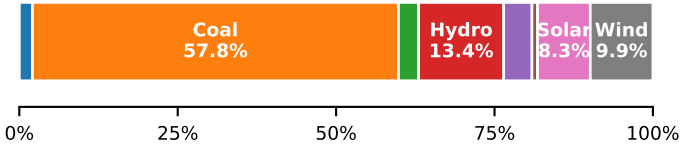


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0207	-1.063
Coal	0.5777	+2.267
Gas	0.0318	-0.633
Hydro	0.1343	+0.808
Nuclear	0.0447	-0.292
Other Fossil	0.0088	-1.916
Solar	0.0832	+0.329
Wind	0.0988	+0.501



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

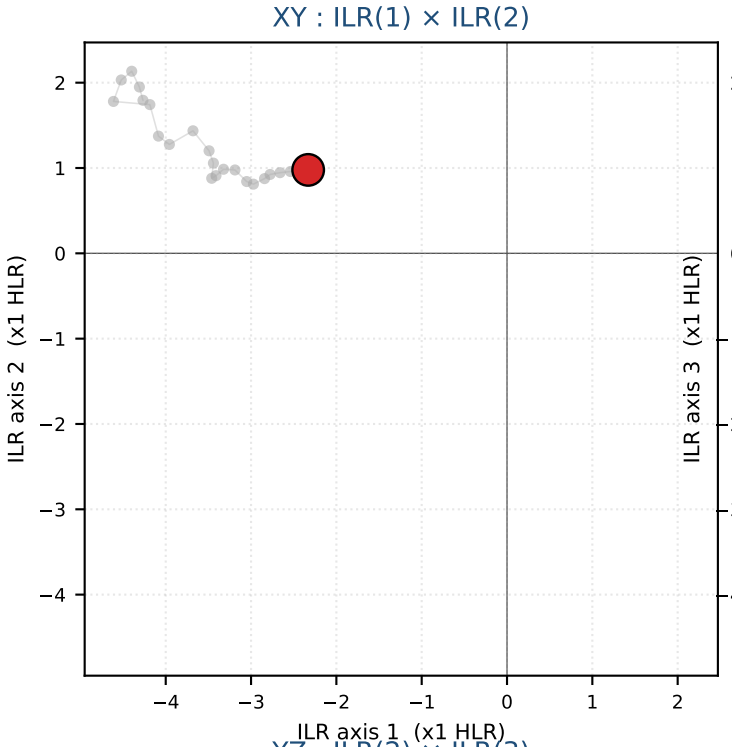
display factor: x1 (raw HLR x 1)

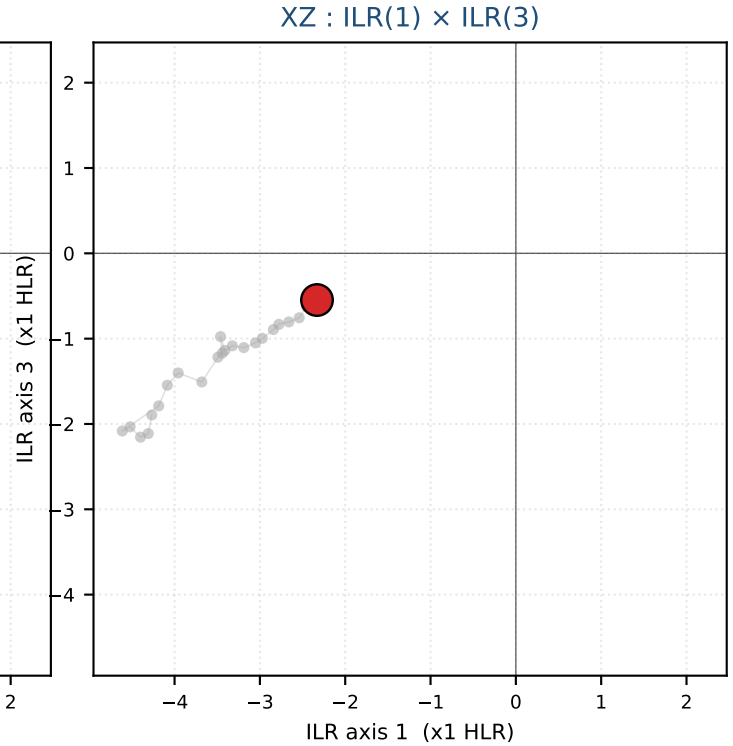
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=25 | 2025 | D=8 T=26 | level 1 first-principles

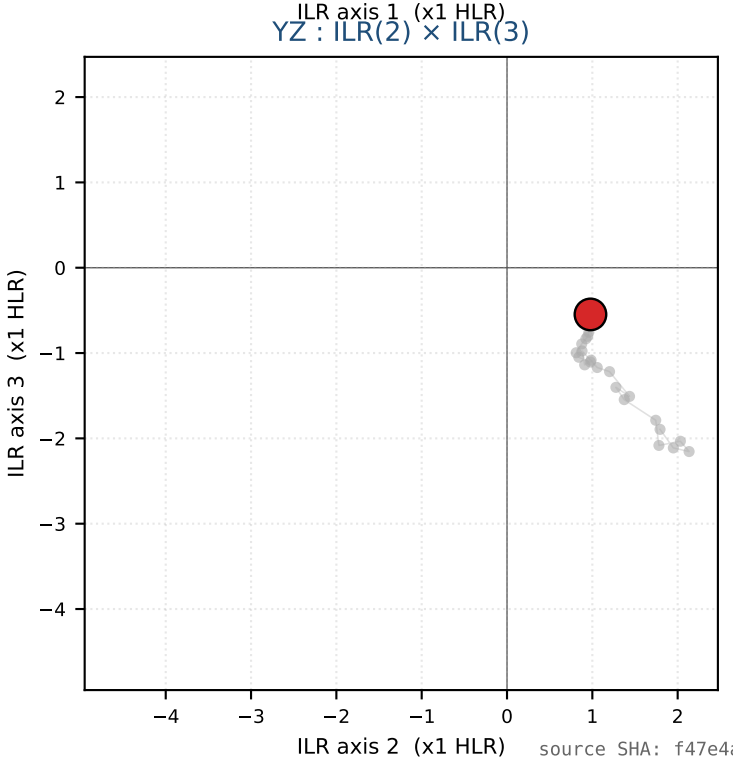
ember_chn | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.4

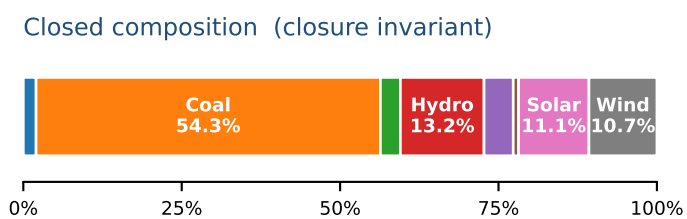




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0201	-1.121
Coal	0.5435	+2.175
Gas	0.0316	-0.671
Hydro	0.1320	+0.760
Nuclear	0.0461	-0.293
Other Fossil	0.0085	-1.988
Solar	0.1110	+0.586
Wind	0.1073	+0.552





closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

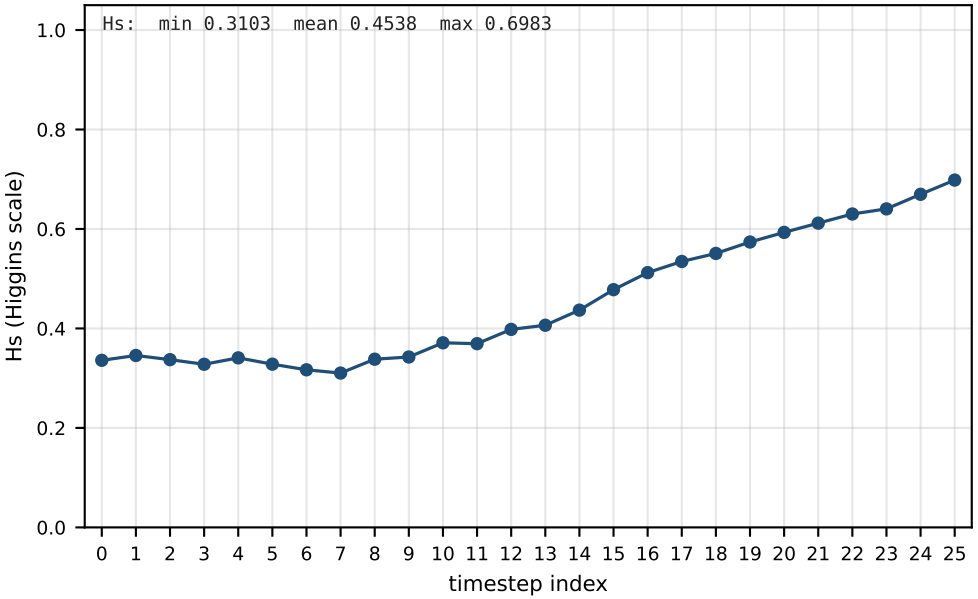
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

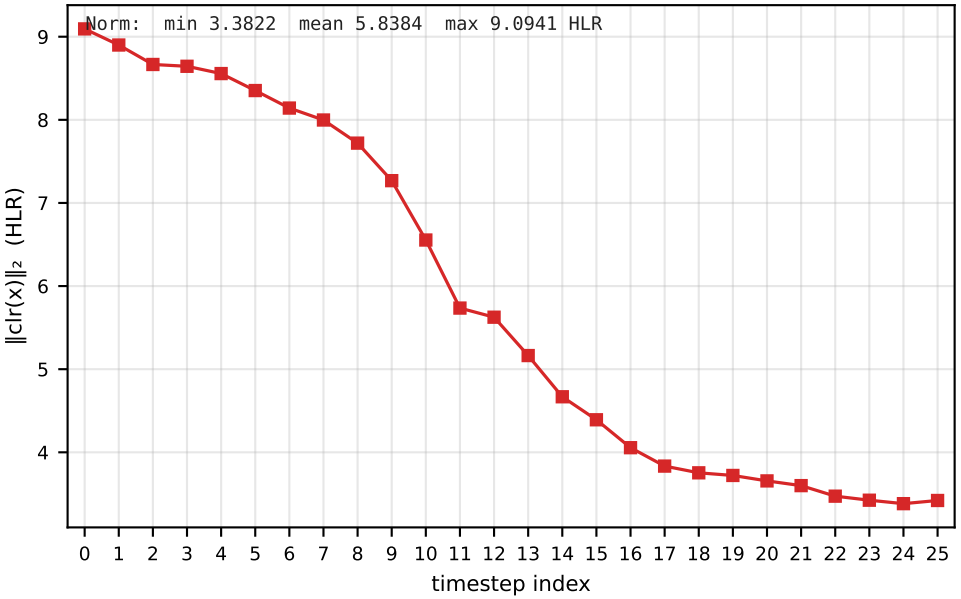
STAGE 1 SUMMARY | ember_chn | D=8 T=26 | Order 1 (first-principles closure)

Engine cnt 2.0.4 | Schema 2.1.0 | Generated 2026-05-06T01:24:03

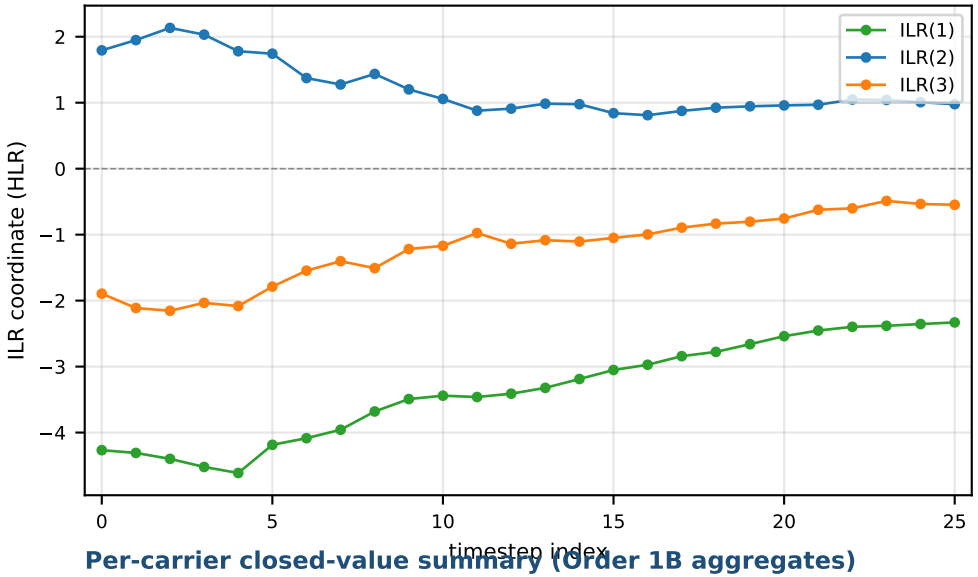
Hs trajectory — single-composition entropy / ln(D)



Aitchison norm — distance from centroid (HLR)



ILR axes 1–3 trajectories (Helmert basis)



Order-1 invariants (whole trajectory)

```
closure invariant     :  max |Σ closed − 1| = 3.33e-16
closure status        :  OK ✓

Hs range              :  [0.3103, 0.6983]
Aitchison norm range  :  [3.3822, 9.0941] HLR
ILR(1) range          :  [-4.6126, -2.3305]
ILR(2) range          :  [+0.8107, +2.1340]
ILR(3) range          :  [-2.1544, -0.4883]

Source file           :  ember_CHN_China_generation_TWh.csv
Source SHA-256        :  f47e4af61ed44a1760162a1350b1375c...
Closed-data SHA-256   :  26dca82d7386e84bb43b3ea16e91ac17...
Content SHA-256       :  4c880f444f3b8c1608cfecf214ee15b3...
Ordering              :  temporal=True, method=by-time
Zero replacement δ    :  1e-15
```

Per-carrier closed-value summary (Order 1B aggregates)

#	carrier	min	mean	max	std
0	Bioenergy	0.0011	0.0091	0.0210	0.0069
1	Coal	0.5435	0.7152	0.8096	0.0781
2	Gas	0.0025	0.0199	0.0336	0.0115
3	Hydro	0.1296	0.1630	0.1917	0.0170
4	Nuclear	0.0118	0.0289	0.0477	0.0126
5	Other Fossil	0.0033	0.0126	0.0349	0.0097
6	Solar	0.0000	0.0182	0.1110	0.0287
7	Wind	0.0004	0.0332	0.1073	0.0344